

# Refinements to Rceattle for eastern Bering Sea pollock

Rceattle 5.8.1 working paper for the 2024 assessment configuration

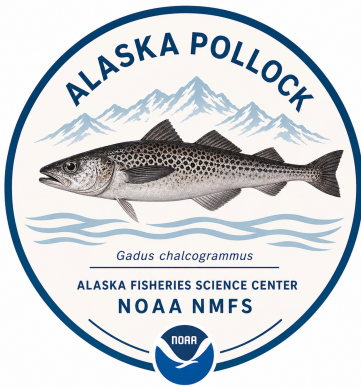
James Ianelli

Grant Adams

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## Abstract

The custom AD Model Builder (ADMB) stock assessment model developed for eastern Bering Sea walleye pollock has a long history of application. For a variety of reasons, including general recommendations from the 2025 Center for Independent Experts (CIE) reviews, the custom ADMB code has been successfully ported to R Template Model Builder (RTMB). That conversion provides access to advanced RTMB features and makes the underlying code easier to understand and modify. Similarly, adopting more generic platforms, such as Rceattle, can provide access to more modern software developments. In this study, we tailored Rceattle code and configurations so that Rceattle can closely mimic the custom ADMB/RTMB assessment code. The Rceattle configuration uses the supported AFSC multinomial composition likelihood and a non-parametric selectivity form. We also evaluated a two-dimensional, first-order autoregressive (AR1) selectivity sensitivity; it did not converge and remains a research topic. An exploratory dynamic structural equation modeling (DSEM) analysis shows how recruitment covariates can be compared by cohort, separately from the final assessment configuration.



Rceattle WORKING PAPER

## 1 Executive Summary

Building from the ADMB-to-RTMB conversion, this analysis evaluates whether Rceattle can be configured to reproduce the main features of the 2024 eastern Bering Sea pollock “PM” assessment while providing a model that is easier to extend. This 5.8.1 update focuses on the Rceattle configuration, comparison with the modified ADMB model, model diagnostics, and projection inputs.

When the Rceattle parameters and fishery selectivity are set to the same values used in the ADMB model, Rceattle reproduces numbers-at-age, spawning biomass, and predicted catch to numerical precision. Mean absolute differences are 0.000117% for spawning biomass and 0.000143% for predicted catch. This result establishes agreement in the population dynamics when both implementations use the same parameter values.

When Rceattle was fit to the data, it reached OK convergence status with a positive-definite Hessian. The least-identified direction remained in annual ATS/AVO selectivity deviations, but the added direct coefficient-increment penalty reduced the condition number below the warning threshold. Relative to the ADMB fit, mean absolute differences were 1.4% for spawning biomass, 0.7% for recruitment, and 10.3% for total biomass. These small differences are concentrated in the earliest years; the remaining uncertainty is primarily estimability of the annual selectivity deviations.

The 2D age-by-year AR1 selectivity sensitivity was included as an initial diagnostic attempt. Results appeared broadly similar to the non-parametric constrained selectivity fits. However, because some parameters reached bounds, the model failed to converge. Closer examination of this selectivity form will continue as a research topic.

One-step-ahead residuals and nine terminal-year-consistent retrospective peels have been generated from the Rceattle configuration. Aggregate composition residual dispersion is lower than the standard-normal reference. Mohn's rho is 0.261 for spawning biomass, 0.119 for total biomass, 0.147 for recruitment, and -0.077 for fishing mortality. Forty-eight of 50 jitters returned to the best solution; two found a substantially poorer mode. These results focus subsequent work on selectivity identification, retrospective bias, and observation-model sensitivity.

## 2 Purpose and Scope

The September 2026 review focuses on whether the Rceattle configuration has sufficient documentation for Plan Team evaluation. This working paper serves that focused purpose rather than the broader purpose of a full Stock Assessment and Fishery Evaluation (SAFE) chapter.

This working paper therefore records:

- what was run from those development files;
- how the Rceattle outputs compare with the modified ADMB reference;
- which diagnostics are available now;
- which sensitivities remain research-only; and
- what work would help structure the next evaluation pass.

The report uses the [Alaska Stock Assessment Report \(asar\) package](#) for glossary conventions and accessibility guidance ([Schiano et al. 2026](#)). Those practices helped prepare the Quarto source for Section 508 review, including semantic headings and tables, descriptive figure captions, and alternative text. They support accessibility but do not replace final manual and agency-approved compliance checks.

Standard SAFE report sections such as background material, fishery history, ecosystem narrative, reference-point tables, and harvest advice are outside the scope of this document. Projection calculations are included only to test the export and use of the Rceattle results.

## 3 Rceattle Configuration

### 3.1 Model Inputs

The Rceattle configuration uses the prepared workbook `Data/EBS_24_pollock_m23_rceattle_full_1964-2024.xlsx`. The data builder retains every age from 1 through 15. Fishery compositions use nominal sample sizes; bottom trawl survey (BTS) and acoustic-trawl survey (ATS) sample sizes use the integer values aligned with the modified ADMB model. Age-1 survey selectivity remains separately parameterized from the age-2–15 curve, and each likelihood contribution evaluates the complete 15-age composition. Compositions use the supported `MultinomialAFSC` likelihood. We added a robustness constant of 0.001 to both observed and expected proportions before evaluating the likelihood. This prevents  $\log(0)$  and limits the influence of observations in bins with expected proportions near zero.

The run uses Rceattle version 5.8.1 from upstream commit 7bb694788012, with the model configuration synchronized to `Rceattle-models` commit 3dc99ddfbf9e.

The next-assessment update script review is complete, and execution awaits complete new-year observations. The script duplicates terminal records and places placeholders for new catch, composition, and survey observations. The script serves as a template until complete new-year observations are available; after population, it creates a complete assessment workbook.

### 3.2 Selectivity

The configuration uses **NonParametricPM** fishery selectivity with annual deviations. When fit to the data, the model reached a small maximum gradient. The Hessian condition number was approximately 489,000 and received **OK** status. Its least-identified direction remained concentrated in **sel\_coff\_dev**, specifically the shared ATS/AVO block rather than the shared fishery/CPUE block.

The time-varying selectivity penalties use explicit standard deviations on log-selectivity random-walk increments. The fishery and its shared CPUE selectivity block use an SD of 0.500, ATS and AVO use 0.138, and BTS uses 1.000. ATS and AVO also use a direct coefficient-increment penalty (**Sel\_curve\_pen3** = 0.1) to constrain the across-age common-mode direction that is removed by annual selectivity normalization. The fishery specification also retains the ADMB-matched curve-shape penalty weights (12.5, 1/60, and 1) and an average-selectivity centering weight of 10. These values are penalty coefficients rather than SDs.

The fitting sequence, composition-likelihood sensitivity, and 2D age-by-year AR1 development work are documented in the appendix.

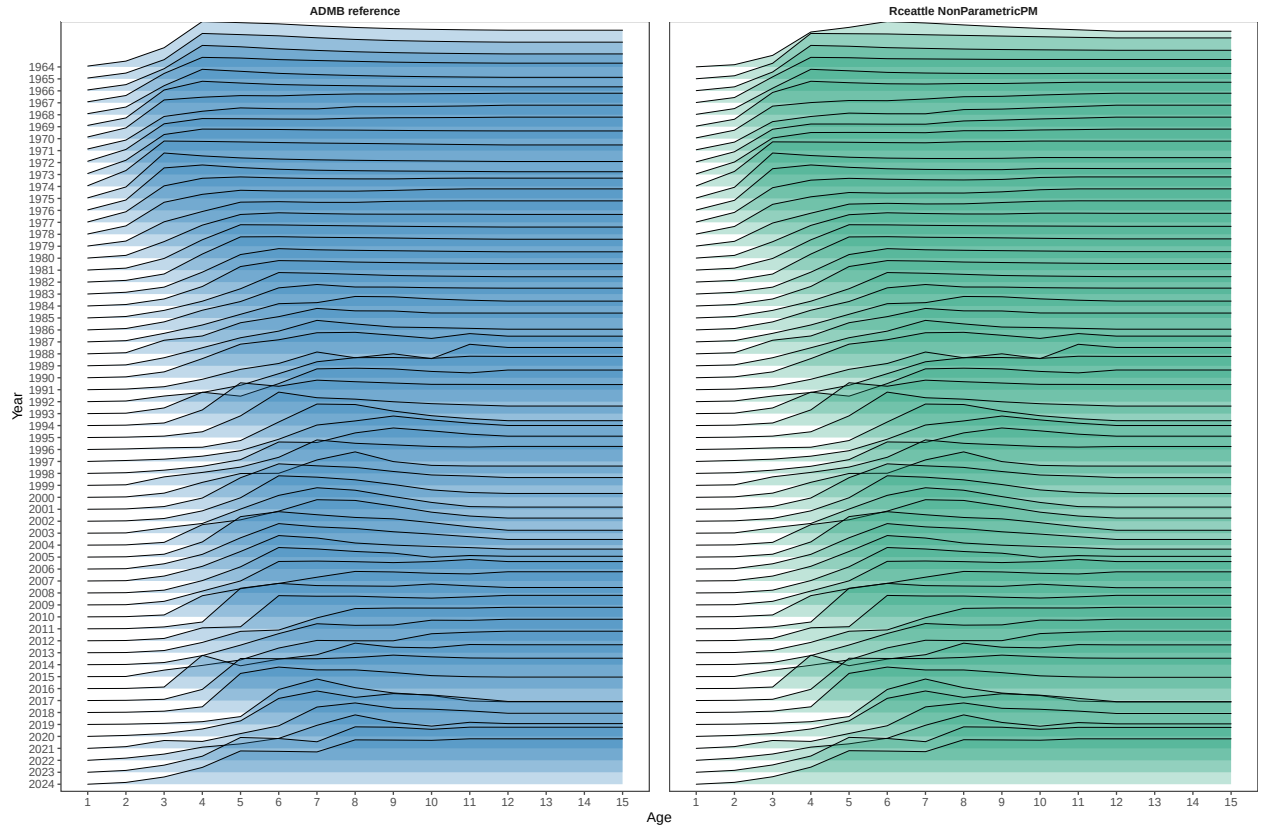


Figure 1: Relative fishery selectivity by year for the ADMB model and Rceattle configuration. Each annual curve is normalized to a maximum of one.

## 4 ADMB Comparison Results

When the Rceattle parameters and fishery selectivity are set to the ADMB values, the two programs give nearly identical population calculations. Mean absolute differences are 0.000117% for spawning biomass and 0.000143% for predicted catch. Annual Rceattle-to-ADMB ratios for both quantities remain extremely close to one (Figure 2).

The reported Rceattle likelihood contributions are 140.191 for indices, -123.761 for catch, 3572.874 for age compositions, and 98.379 for recruitment deviations. Their sum is 3704.588. The ADMB `tot_like` and `dat_like` values are 736.724 and 459.931, respectively, but cannot be compared directly because the two programs include different likelihood terms, penalties, and constants. The ADMB maximum absolute gradient is 3.47e-04; no Rceattle model-parameter gradient is available from this comparison because Rceattle does not estimate parameters in this run.

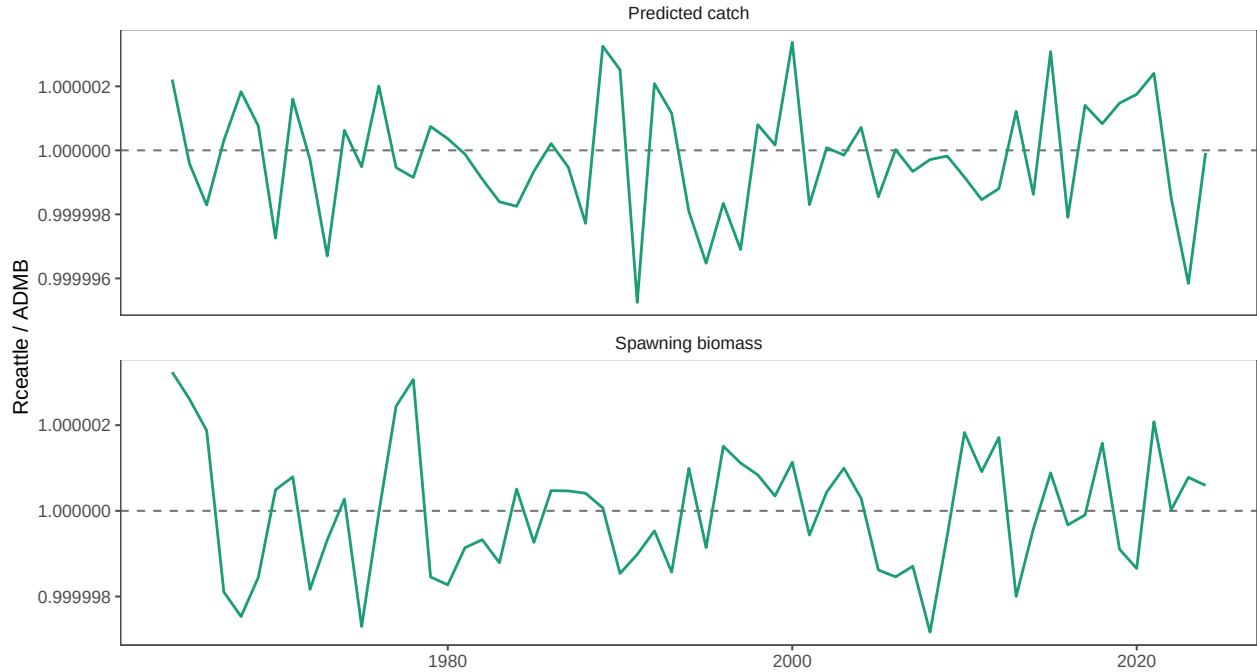


Figure 2: Rceattle-to-ADMB ratios for spawning biomass and predicted catch when Rceattle uses the ADMB parameter values.

Table 1: Agreement between fitted Rceattle trajectories and the modified ADMB reference. Correlation is the Pearson correlation across annual estimates. Mean absolute (%) and maximum absolute (%) are the average and largest absolute annual percentage differences, respectively. The 2024 difference (%) is the signed terminal-year difference of Rceattle relative to ADMB; negative values indicate that the Rceattle estimate is lower.

| Quantity               | Correlation | Mean absolute (%) | Maximum absolute (%) | 2024 difference (%) |
|------------------------|-------------|-------------------|----------------------|---------------------|
| Rceattle configuration |             |                   |                      |                     |
| Recruitment            | 1.00        | 0.70              | 10.28                | -6.45               |
| Spawning biomass       | 1.00        | 1.39              | 10.01                | 0.06                |
| Total biomass          | 0.98        | 10.32             | 95.63                | 2.61                |

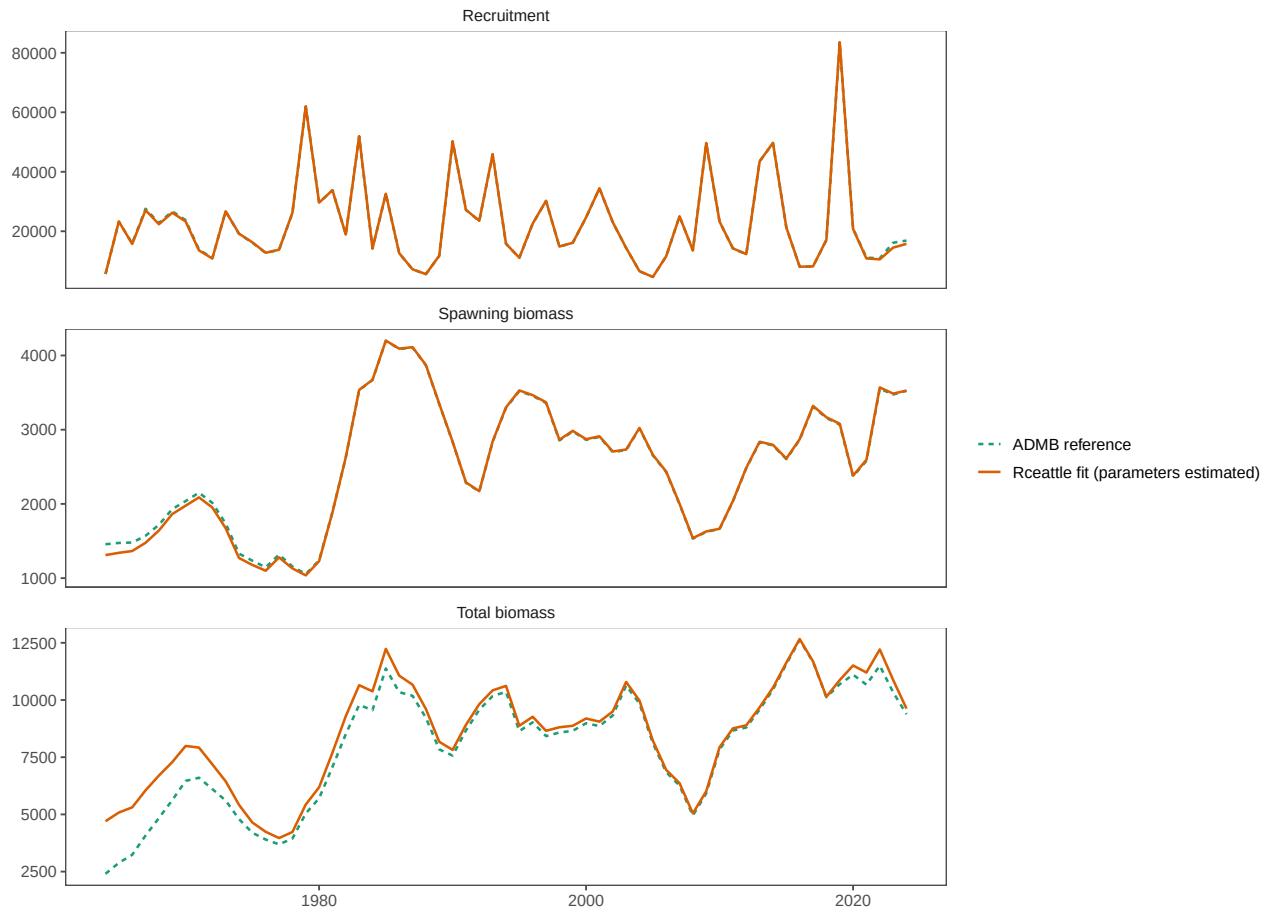


Figure 3: ADMB and fitted Rceattle trajectories for spawning biomass, recruitment, and total biomass. The Rceattle series is from the fit in which its parameters were estimated.

When Rceattle estimates its parameters, the fitted trajectories differ most from ADMB during the early period and in the terminal recruitment estimate (Figure 3). These trajectories are distinct from the run in which Rceattle parameters are set to the ADMB values. The fitted trajectories remain highly correlated with the ADMB reference, with relatively small mean annual differences (Table 1). Development sensitivities are documented in the appendix.

The Rceattle fit has a small maximum gradient and a positive-definite Hessian. Adding the direct ATS/AVO coefficient-increment penalty reduced the condition number from approximately 1.14 million to 489,000 and removed the prior Hessian-conditioning warning. The residual weakest direction remains almost entirely in the shared time-varying ATS/AVO selectivity coefficients, not the fishery/CPUE coefficients, so this block remains an appropriate focus for sensitivity checks even though the formal warning is resolved.

## 5 Diagnostics

The numerical convergence checks for the fitted Rceattle model, including the objective, maximum gradient, and Hessian status, indicated convergence (Table 2). The fit and status of the other diagnostic products are provided in Table 3.

Table 2: Stored Rceattle convergence checks for the evaluated configurations.

| Model        | Status | Objective  | Max<br>gradient | Weakest<br>component | PD<br>Hessian |
|--------------|--------|------------|-----------------|----------------------|---------------|
| Rceattle fit | OK     | 716.948070 | 0.000041        | sel_inf              | TRUE          |

Table 3: Diagnostic status for the September 2026 review.

| Diagnostic                                | Current status                                                                                                                                                             | Review use                                                                                                                                                                            |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADMB-parameter comparison<br>Rceattle fit | Completed and saved in the ADMB-to-Rceattle comparison outputs.<br>Completed with OK status, a positive-definite Hessian, and a condition number of approximately 489,000. | Primary numerical agreement check.<br><br>Primary Rceattle comparison fit to the data.                                                                                                |
| 2D AR1 selectivity sensitivity            | Received a FAIL convergence status, an unsuccessful Hessian inversion, and parameter-bound warnings.                                                                       | Requires further research before it can be considered as an alternative selectivity form.                                                                                             |
| OSA residuals                             | Completed for 1,864 residuals and saved in RDS outputs.                                                                                                                    | Useful, but analytical and exactly reproduced index groups need interpretation.                                                                                                       |
| Rceattle retrospectives                   | Nine peels completed for the Rceattle fit with OK status.                                                                                                                  | The revised peels preserve data availability and fix terminal fishery/CPUE selectivity increments at zero, holding the terminal selectivity-at-age curve at its preceding-year value. |
| Legacy ADMB retrospectives                | Existing ADMB retro.rep files are present.                                                                                                                                 | Provides historical context distinct from the Rceattle retrospective peels.                                                                                                           |
| Rceattle 5.8.1 self-tests                 | Fifty initial-start and 50 estimated-start refits completed with OK status.                                                                                                | Supports the revised positive-index simulation workflow; it does not resolve observed-data multimodality.                                                                             |
| Jitters and likelihood profile            | Forty-eight of 50 jitters returned to the best mode; two WARN fits reached an inferior mode. The age-3+ M profile was shallow with its grid minimum at 0.20.               | Documents multimodality and weak information about natural mortality.                                                                                                                 |

Table 4: Rceattle 5.8.1 self-test and paired-restart validation for the upstream-supported EBS configuration.

| Metric                                                             | Value        |
|--------------------------------------------------------------------|--------------|
| Initial-start self-tests converged                                 | 5.000000e+01 |
| Estimated-start self-tests converged                               | 5.000000e+01 |
| Paired fits with objective difference greater than 0.0001          | 0.00000e+00  |
| Maximum absolute paired objective difference                       | 3.09564e-08  |
| Median across-simulation median SSB percent bias                   | -1.055%      |
| Median terminal-year SSB percent bias                              | -0.907%      |
| Fleets with at least one row above 2 percent rejection probability | 2.00000e+00  |
| Maximum row-level non-positive probability                         | 6.66814e-02  |

Table 5: Fleet-level probability of a non-positive index draw under the fitted, untruncated observation distribution.

| Fleet | Code | Distribution | Rows | Maximum row probability | Mean row probability | Rows above 2 |
|-------|------|--------------|------|-------------------------|----------------------|--------------|
| ATS   | 4    | Lognormal    | 19   | 0.00%                   | 0.00%                |              |
| ATS_1 | 6    | Lognormal    | 18   | 0.00%                   | 0.00%                |              |
| AVO   | 2    | Normal       | 18   | 3.88%                   | 0.23%                |              |
| BTS   | 3    | MVN          | 42   | 6.64%                   | 0.17%                |              |
| BTS_1 | 5    | Lognormal    | 42   | 0.00%                   | 0.00%                |              |
| CPUE  | 7    | Normal       | 12   | 0.00%                   | 0.00%                |              |

## 5.1 Rceattle 5.8.1 self-test validation

Rceattle 5.8.1 changed simulation of natural-scale normal and multivariate normal survey indices. The fitted likelihood remains an untruncated normal; the simulator redraws non-positive values because `data_check()` excludes them. The resulting self-test therefore uses a normal distribution truncated at zero whenever rejection occurs.

The ADMB-to-Rceattle comparison was rerun first with Rceattle 5.8.1. Its saved summary gives mean absolute differences of 0.000117% for spawning biomass and 0.000143% for predicted catch. This confirms that the package update preserved the agreement in the population calculations.

The self-test uses the same delivered EBS workbook and supported `MultinomialAFSC` composition criterion as the Rceattle configuration. The observed-data fit reproduced the stored objective of 716.948070, maximum gradient of 0.000041, and OK status with a positive-definite Hessian. The Hessian condition number was approximately  $4.9 \times 10^5$ ; its least-identified direction remained concentrated in the shared ATS/AVO `sel_coff_dev` block rather than fishery/CPUE.

All 50 phased simulations converged with OK status when refitted from the original initial values. Repeating the same simulated datasets from the fitted parameter values also produced 50 OK fits. Paired objectives agreed within  $3.1 \times 10^{-8}$ , so these simulations showed no start-dependent mode. Median spawning-biomass bias was -1.06% across years and -0.91% in the terminal year. These results validate the revised self-test workflow; they do not rule out the mode previously encountered when fitting the observed data.

One AVO row had a 3.76% non-positive probability and one BTS multivariate-normal row had a 6.67% probability. Those rows exceed the package's 2% warning threshold and should be identified when interpreting recovery. CPUE and the remaining index series had negligible or zero marginal rejection probability.



## 5.2 OSA Residuals

OSA residuals run on the Rceattle configuration. The aggregate composition fits are generally close, while all three OSA quantile plots are narrower than the standard-normal reference (Figure 4). These diagnostics apply to the fitted `NonParametricPM` selectivity configuration. The failed 2D-AR1 sensitivity and ADMB-to-Rceattle calculation use separate results. The composition diagnostics use `afscOSA` version 0.0.1 and pass Rceattle’s internally calculated residuals to `afscOSA::run_osa()`. The residual builder uses complete ages 1–15, with nominal fishery sample sizes and the integer BTS and ATS sample sizes aligned with the modified ADMB model. This follows the current [afscOSA interpretation guidance](#) while retaining Rceattle’s residual calculation.

Following that guidance, the aggregate age-composition fits are the first check for systematic misfit, including patterns that could indicate an unsuitable selectivity form. The red vertical ranges contain 95% of data simulated from the fitted multinomial model conditional on the aggregate sample size and estimated parameter values held constant. They represent conditional simulation variability. The quantile–quantile plots then assess whether the OSA residuals resemble a standard normal distribution. The standard deviation of normalized residuals (SDNR) should be near one, and the displayed tail statistics provide a separate check on extreme residual behavior. Finally, the OSA and Pearson bubble plots are used to look for age, year, or cohort patterns that may suggest the source of misfit. Each statistic and nominal interval contributes to an integrated diagnostic interpretation rather than an automatic model-rejection rule.

The aggregate observed and fitted compositions are generally close, although the survey panels contain some age-specific departures. The OSA distributions are narrower than the standard-normal reference for all three composition sources: SDNR is 0.64 for the fishery, 0.80 for BTS, and 0.53 for ATS. Their tail diagnostics also show shorter-than-expected tails. Together these results indicate underdispersion relative to the fitted multinomial observation model. Model selection requires this information alongside the full suite of diagnostics. The bubble plots should instead be used to identify persistent age, year, or cohort patterns and to guide focused checks of selectivity, composition weighting, and observation-model assumptions. One Pearson residual (8.86) is capped at 6 in the figure, following the package’s plotting convention.

## 5.3 Retrospective patterns

Nine retrospective peels were run from the Rceattle fit. The unpeeled model ends in 2024, and the peeled fits end in 2023 through 2015. Each peel preserves the data-availability pattern of the 2024 model. Thus, fishery age compositions end one year before the peel terminal year, survey series retain their historical sampling schedule and terminal lags, and earlier discontinued series retain their original ending years.

Each peel applies the fitting sequence documented in the appendix. The terminal-year fishery and mirrored catch-per-unit-effort selectivity increments are fixed at zero. All nine final fits received OK convergence status, with maximum gradients from 2.1e-05 to 6.4e-05.

The `NonParametricPM` fishery selectivity is parameterized at age. The base `sel_coff` parameters and annual `sel_coff_dev` random-walk increments apply to age bins; the fishery uses 12 estimated bins and holds older ages at the last estimated value. Fixing every terminal-year increment at zero therefore holds the realized terminal selectivity-at-age curve at the preceding-year curve. Across all peels and both the fishery and mirrored catch-per-unit-effort series, the maximum absolute terminal-to-preceding-year difference was 8.9e-16. These Rceattle fits are separate from the legacy AD Model Builder (ADMB) retrospective files.

The estimated Mohn’s rho values are 0.261 for spawning biomass, 0.119 for total biomass, 0.147 for recruitment, and -0.077 for fishing mortality. The positive biomass and recruitment values indicate that the peeled models generally estimate higher quantities at their terminal years than the corresponding unpeeled estimates. The negative fishing-mortality value indicates lower terminal fishing mortality in the peels. Spawning biomass shows the largest average relative difference. The consistent OK statuses and small gradients support interpretation of these values as properties of the revised retrospective specification, while the magnitude of spawning-biomass rho motivates continued evaluation of selectivity, survey influence, and recent recruitment estimates.

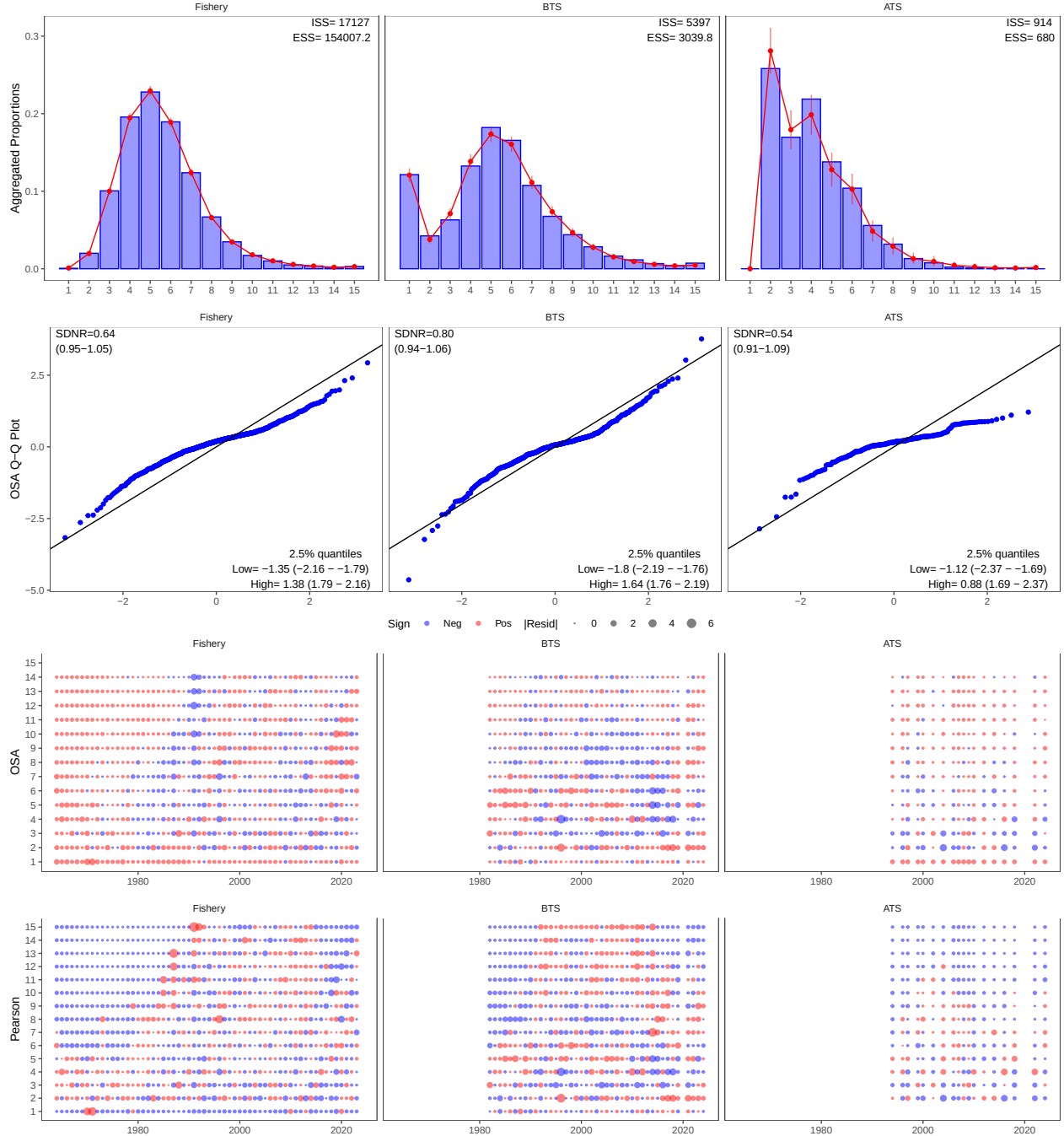


Figure 4: Age-composition diagnostics for the Rceattle fit. The fit uses complete ages 1–15, nominal fishery sample sizes, and BTS and ATS integer sample sizes aligned with the modified ADMB model. Diagnostics were produced with afscOSA for the fishery, bottom trawl survey (BTS), and acoustic-trawl survey (ATS). The top row compares aggregate observed proportions (blue bars) with fitted proportions (red points and lines); red vertical ranges contain 95% of data simulated conditionally from the fitted multinomial model, aggregate sample size, and estimated parameter values held constant. The second row shows quantile–quantile plots of Rceattle’s internal one-step-ahead (OSA) residuals with the standard deviation of normalized residuals (SDNR), its expected 95% interval, and lower and upper tail diagnostics. The lower rows show OSA and Pearson residuals by age and year; red and blue indicate positive and negative residuals, and bubble area represents magnitude. Absolute residuals greater than 6 are capped at 6 for display. The fitted model retains ATS 2020; its sample size of one yields zero rounded counts in the afscOSA plotting conversion, so the plotting layer omits that row.

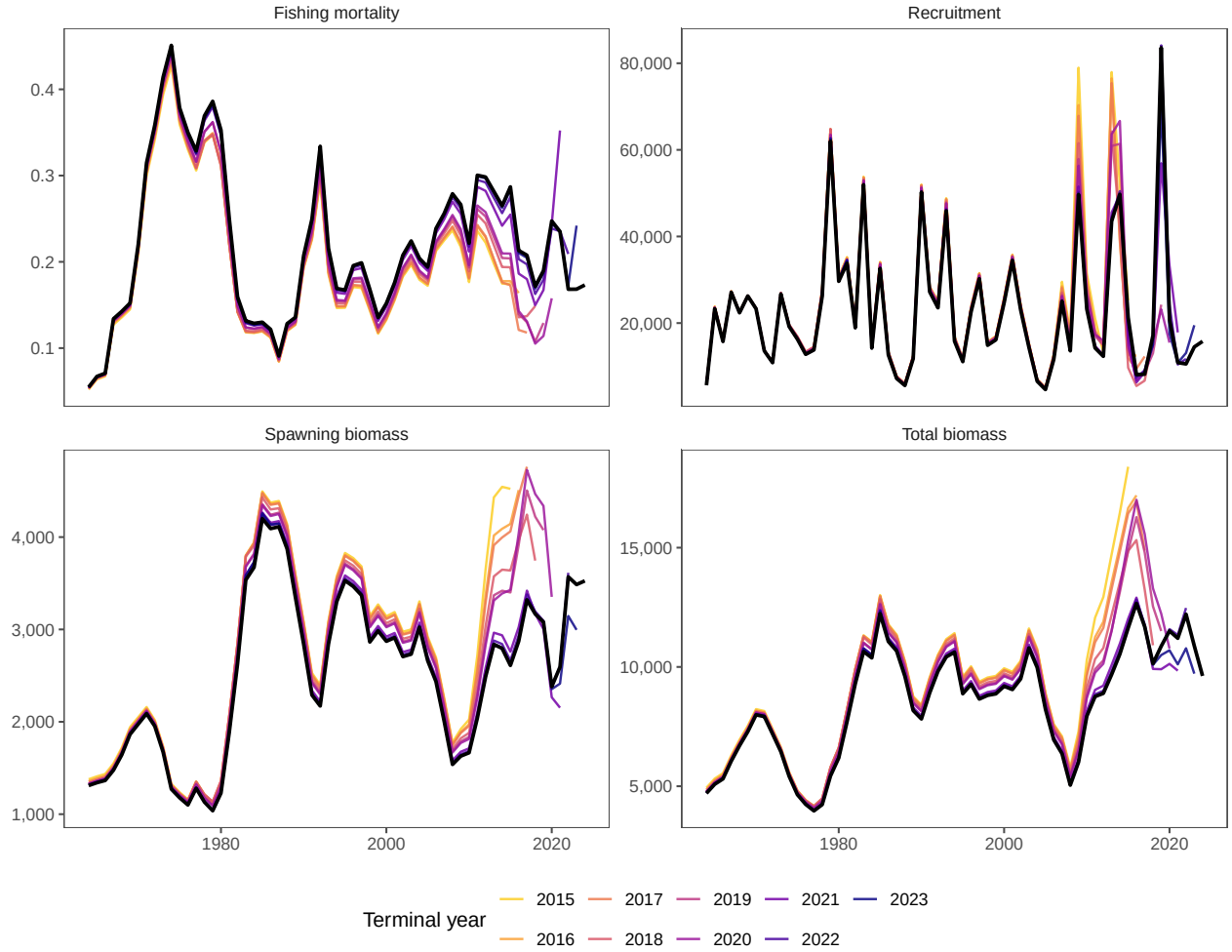


Figure 5: Nine retrospective peels for the Rceattle configuration. The black line is the unpeeled model ending in 2024; colored lines are successive peels ending in 2023 through 2015 and are plotted through each peel's terminal year. Each peel preserves the original data-availability lags and fixes the terminal fishery/CPUE selectivity-at-age increment at zero, holding the terminal curve at its preceding-year value. All nine fits received OK convergence status.

Table 6: Mohn's rho for the nine retrospective peels of the Rceattle configuration. Values are the mean relative difference between a peeled estimate in its terminal year and the corresponding estimate from the unpeeled model. Each peel preserves the 2024 data-availability pattern and fixes the terminal fishery/CPUE selectivity-at-age increment at zero.

| Quantity          | Peels | Mohn's rho |
|-------------------|-------|------------|
| Total biomass     | 9     | 0.119      |
| Spawning biomass  | 9     | 0.261      |
| Recruitment       | 9     | 0.147      |
| Fishing mortality | 9     | -0.077     |

## 6 Projections

The saved Rceattle fit supplies every population and fishery input for the Standard Projection Model calculations implemented in `spmR` (J. Ianelli 2026). The run uses `spmR` version 0.3.0 from pinned main-branch commit `e86fc6aa`. The adapter in `R/write_spmr_projection_inputs.R` writes terminal abundance, natural mortality, maturity, spawning and fishery weights, the arithmetic mean of fishery selectivity-at-age over the most recent five assessment years (2020–2024), recruitment, and spawning-biomass histories to `pm.prj`. The same five years define the recent average fishing mortality. The reproducible runner `R/run_spmr_tier3_projection.R` then applies those inputs to all seven Tier 3 alternatives. The report reads the saved projection products, so every table and figure in this section comes from the same saved Rceattle fit. The complete field mapping and validation sequence are documented in Appendix A-2.

Table 7: Projection files generated from the saved Rceattle fit for the `spmR`/SPM calculations. Checksums show the first eight MD5 characters; the complete values are retained in `manifest.csv`.

### spmR Projection Files

| file                                              | role                                                       | exists | checksum |
|---------------------------------------------------|------------------------------------------------------------|--------|----------|
| <code>spm.dat</code>                              | SPM setup                                                  | TRUE   | 8b4f5e91 |
| <code>pm.prj</code>                               | Rceattle-derived species input                             | TRUE   | e91692f1 |
| <code>tacpar.dat</code>                           | TAC parameters                                             | TRUE   | 3b02746e |
| <code>spm</code>                                  | SPM executable                                             | TRUE   | 6f52316e |
| <code>age_schedules.csv</code>                    | Rceattle-derived age schedules                             | TRUE   | 3c450f47 |
| <code>fishery_selectivity_recent_years.csv</code> | Annual fishery selectivity used in the recent-year average | TRUE   | 0377c30e |

Table 8: Key settings and source information for the Rceattle-derived SPM projection.

### Generated SPM Setup and Source Information

| Item                               | Value                                                                                                                                          |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Rceattle source fit                | <code>nonparametric_pm</code>                                                                                                                  |
| Source checksum                    | 334f8c14b918536c53e4abaa0fc116ae                                                                                                               |
| Composition likelihood             | MultinomialAFSC                                                                                                                                |
| Composition sample sizes           | fishery nominal; BTS and ATS integer sample sizes aligned with modified ADMB                                                                   |
| <code>spmR</code> version          | 0.3.0                                                                                                                                          |
| <code>spmR</code> commit           | <code>e86fc6aa</code>                                                                                                                          |
| Terminal assessment year           | 2024                                                                                                                                           |
| Fishery selectivity rule           | arithmetic mean of annual selectivity-at-age                                                                                                   |
| Fishery selectivity years          | 2020, 2021, 2022, 2023, 2024                                                                                                                   |
| Recent average-F years             | 2020, 2021, 2022, 2023, 2024                                                                                                                   |
| Spawning timing                    | 0.25 year from January 1 (start of April)                                                                                                      |
| Rceattle encoding                  | <code>spawn_month</code> = 3 elapsed months; $3/12 = 0.25$                                                                                     |
| SPM encoding                       | <code>Spawnmo</code> = 4 as a one-based calendar-month code; $(4 - 1)/12 = 0.25$                                                               |
| Timing compatibility check         | Matched: both calculations apply survival through 0.25 of the year                                                                             |
| Spawning-biomass timing convention | Mature spawning biomass after survival through 0.25 of the year: $N \times \exp(-0.25 Z) \times \text{maturity} \times \text{spawning weight}$ |
| Projection begin year              | 2025                                                                                                                                           |
| Projection years                   | 14                                                                                                                                             |
| Simulations                        | 1000                                                                                                                                           |
| Alternatives                       | 1, 2, 3, 4, 5, 6, 7                                                                                                                            |

Table 9: SPM projection alternatives included in the generated setup file.

## SPM Projection Alternatives

| Alternative | Name                        | Description                                                                                   |
|-------------|-----------------------------|-----------------------------------------------------------------------------------------------|
| 1           | Maximum permissible ABC     | Maximum permissible Tier 3 ABC harvest rate.                                                  |
| 2           | Author-specified ABC        | Author-specified adjustment to the Tier 3 ABC harvest rate; the current multiplier is 1.0.    |
| 3           | Average recent F            | Recent average fishing mortality from the Rceattle-derived input file.                        |
| 4           | Alternative SPR rate        | User-specified SPR rate of 0.60.                                                              |
| 5           | No fishing                  | Zero fishing mortality after the fixed-catch years.                                           |
| 6           | OFL threshold determination | OFL harvest rate for threshold and status-determination calculations.                         |
| 7           | Status-determination ramp   | Maximum permissible ABC for three years followed by the OFL status-determination calculation. |

### 6.1 Seven Tier 3 Scenario Results

Catches for 2025 and 2026 are fixed at 1,350 thousand t in every alternative; the alternatives therefore first diverge in 2027 (Table 10). The reported values are simulation means from 1,000 SPM simulations.

Table 10: Tier 3 projection results for all seven FMP alternatives using inputs generated from the Rceattle EBS pollock model. Catch and biomass quantities are in thousand t;  $B/B_{35\%}$  is spawning biomass relative to the Tier 3  $B_{35\%}$  proxy.

## Seven Tier 3 Projection Alternatives

| Alt | Scenario                    | Year | Catch | ABC   | OFL   | SSB   | F     | B/B35 |
|-----|-----------------------------|------|-------|-------|-------|-------|-------|-------|
| 1   | Maximum permissible ABC     | 2027 | 1,864 | 1,864 | 2,259 | 2,476 | 0.350 | 1.17  |
| 1   | Maximum permissible ABC     | 2028 | 1,308 | 1,308 | 1,583 | 2,115 | 0.310 | 1.00  |
| 2   | Author-specified ABC        | 2027 | 1,864 | 1,864 | 2,259 | 2,476 | 0.350 | 1.17  |
| 2   | Author-specified ABC        | 2028 | 1,308 | 1,308 | 1,583 | 2,115 | 0.310 | 1.00  |
| 3   | Average recent F            | 2027 | 1,132 | 1,132 | 2,259 | 2,582 | 0.200 | 1.22  |
| 3   | Average recent F            | 2028 | 1,015 | 1,015 | 1,963 | 2,446 | 0.200 | 1.16  |
| 4   | Alternative SPR rate        | 2027 | 856   | 856   | 2,259 | 2,619 | 0.150 | 1.24  |
| 4   | Alternative SPR rate        | 2028 | 803   | 803   | 2,103 | 2,585 | 0.150 | 1.22  |
| 5   | No fishing                  | 2027 | 0     | 0     | 2,259 | 2,726 | 0.000 | 1.29  |
| 5   | No fishing                  | 2028 | 0     | 0     | 2,455 | 3,034 | 0.000 | 1.44  |
| 6   | OFL threshold determination | 2027 | 1,561 | 1,561 | 1,561 | 2,017 | 0.370 | 0.95  |
| 6   | OFL threshold determination | 2028 | 1,210 | 1,210 | 1,210 | 1,831 | 0.340 | 0.87  |
| 7   | Status-determination ramp   | 2027 | 1,509 | 1,809 | 1,809 | 2,211 | 0.320 | 1.05  |
| 7   | Status-determination ramp   | 2028 | 1,424 | 1,424 | 1,424 | 1,977 | 0.360 | 0.94  |

Table 12: Candidate ABC projection given assumed future catches. Catch is fixed at 1,300 thousand t; ABC is the maximum permissible ABC calculated by SPM.  $B/B_{35\%}$  is mean spawning biomass relative to the Tier 3 proxy.

## Candidate ABC Projection Given Assumed Future Catches

| Year | Catch | ABC   | OFL   | Mean $B$ | $B/B_{35\%}$ |
|------|-------|-------|-------|----------|--------------|
| 2025 | 1,300 | 2,360 | 2,878 | 3,523    | 167%         |
| 2026 | 1,300 | 2,131 | 2,588 | 3,013    | 143%         |
| 2027 | 1,300 | 1,891 | 2,295 | 2,597    | 123%         |
| 2028 | 1,300 | 1,583 | 1,909 | 2,366    | 112%         |
| 2029 | 1,300 | 1,446 | 1,763 | 2,364    | 112%         |
| 2030 | 1,300 | 1,459 | 1,786 | 2,446    | 116%         |
| 2031 | 1,300 | 1,520 | 1,871 | 2,547    | 120%         |
| 2032 | 1,300 | 1,594 | 1,967 | 2,644    | 125%         |

Table 11: Age-specific schedules applied in SPM. Natural mortality, maturity, and weights come from the terminal Rceattle configuration; fishery selectivity is the arithmetic mean at age over 2020–2024.

## Projection Age-Specific Schedules

| Age | Spawning_weight | Fishery_weight | Maturity | Natural_mortality | Selectivity |
|-----|-----------------|----------------|----------|-------------------|-------------|
| 1   | 0.085           | 0.025          | 0.000    | 0.900             | 0.000       |
| 2   | 0.196           | 0.186          | 0.008    | 0.450             | 0.061       |
| 3   | 0.322           | 0.401          | 0.289    | 0.300             | 0.220       |
| 4   | 0.534           | 0.667          | 0.641    | 0.300             | 0.414       |
| 5   | 0.621           | 0.663          | 0.842    | 0.300             | 0.775       |
| 6   | 0.707           | 0.720          | 0.901    | 0.300             | 1.031       |
| 7   | 0.822           | 0.791          | 0.947    | 0.300             | 1.302       |
| 8   | 0.946           | 0.905          | 0.963    | 0.300             | 1.530       |
| 9   | 1.051           | 1.009          | 0.970    | 0.300             | 1.456       |
| 10  | 1.113           | 1.061          | 1.000    | 0.300             | 1.394       |
| 11  | 1.178           | 1.138          | 1.000    | 0.300             | 1.406       |
| 12  | 1.290           | 1.257          | 1.000    | 0.300             | 1.353       |
| 13  | 1.332           | 1.267          | 1.000    | 0.300             | 1.353       |
| 14  | 1.367           | 1.290          | 1.000    | 0.300             | 1.353       |
| 15  | 1.415           | 1.368          | 1.000    | 0.300             | 1.353       |

### 6.2 Alternative 2 Fixed-Catch Projection

This additional run applies Alternative 2 and fixes catch at 1,300 thousand t for 2025–2032. The projection horizon includes every fixed-catch year.

### 6.3 Projection Figures

The figures summarize the 1,000 simulations for each alternative. Lines show simulation medians and ribbons show central 90% intervals.

Table 13: Mean SPM results by alternative and projection year. Biomass, catch, ABC, and OFL are in thousand t.

## SPM Projection Means

| Alt | Year | B     | Catch | ABC   | OFL   | F     | $B/B_{35\%}$ |
|-----|------|-------|-------|-------|-------|-------|--------------|
| 1   | 2025 | 3,516 | 1,350 | 2,360 | 2,878 | 0.188 | 166%         |
| 1   | 2026 | 2,986 | 1,350 | 2,116 | 2,570 | 0.210 | 141%         |
| 1   | 2027 | 2,476 | 1,864 | 1,864 | 2,259 | 0.354 | 117%         |
| 1   | 2028 | 2,115 | 1,308 | 1,308 | 1,583 | 0.306 | 100%         |
| 1   | 2029 | 2,179 | 1,265 | 1,265 | 1,545 | 0.303 | 103%         |
| 1   | 2030 | 2,321 | 1,365 | 1,365 | 1,671 | 0.311 | 110%         |
| 1   | 2031 | 2,426 | 1,464 | 1,464 | 1,793 | 0.318 | 115%         |
| 1   | 2032 | 2,498 | 1,531 | 1,531 | 1,874 | 0.322 | 118%         |
| 1   | 2033 | 2,532 | 1,564 | 1,564 | 1,913 | 0.323 | 120%         |
| 1   | 2034 | 2,526 | 1,567 | 1,567 | 1,916 | 0.322 | 120%         |
| 1   | 2035 | 2,518 | 1,564 | 1,564 | 1,912 | 0.323 | 119%         |
| 1   | 2036 | 2,504 | 1,554 | 1,554 | 1,900 | 0.322 | 118%         |
| 1   | 2037 | 2,498 | 1,549 | 1,549 | 1,893 | 0.322 | 118%         |
| 1   | 2038 | 2,511 | 1,551 | 1,551 | 1,896 | 0.321 | 119%         |
| 2   | 2025 | 3,516 | 1,350 | 2,360 | 2,878 | 0.188 | 166%         |
| 2   | 2026 | 2,986 | 1,350 | 2,116 | 2,570 | 0.210 | 141%         |
| 2   | 2027 | 2,476 | 1,864 | 1,864 | 2,259 | 0.354 | 117%         |
| 2   | 2028 | 2,115 | 1,308 | 1,308 | 1,583 | 0.306 | 100%         |
| 2   | 2029 | 2,179 | 1,265 | 1,265 | 1,545 | 0.303 | 103%         |
| 2   | 2030 | 2,321 | 1,365 | 1,365 | 1,671 | 0.311 | 110%         |
| 2   | 2031 | 2,426 | 1,464 | 1,464 | 1,793 | 0.318 | 115%         |
| 2   | 2032 | 2,498 | 1,531 | 1,531 | 1,874 | 0.322 | 118%         |
| 2   | 2033 | 2,532 | 1,564 | 1,564 | 1,913 | 0.323 | 120%         |
| 2   | 2034 | 2,526 | 1,567 | 1,567 | 1,916 | 0.322 | 120%         |
| 2   | 2035 | 2,518 | 1,564 | 1,564 | 1,912 | 0.323 | 119%         |
| 2   | 2036 | 2,504 | 1,554 | 1,554 | 1,900 | 0.322 | 118%         |
| 2   | 2037 | 2,498 | 1,549 | 1,549 | 1,893 | 0.322 | 118%         |
| 2   | 2038 | 2,511 | 1,551 | 1,551 | 1,896 | 0.321 | 119%         |
| 3   | 2025 | 3,516 | 1,350 | 1,420 | 2,878 | 0.188 | 166%         |
| 3   | 2026 | 2,986 | 1,350 | 1,283 | 2,570 | 0.210 | 141%         |
| 3   | 2027 | 2,582 | 1,132 | 1,132 | 2,259 | 0.198 | 122%         |
| 3   | 2028 | 2,446 | 1,015 | 1,015 | 1,963 | 0.198 | 116%         |
| 3   | 2029 | 2,569 | 1,001 | 1,001 | 1,950 | 0.198 | 122%         |
| 3   | 2030 | 2,754 | 1,057 | 1,057 | 2,085 | 0.198 | 130%         |
| 3   | 2031 | 2,915 | 1,124 | 1,124 | 2,239 | 0.198 | 138%         |
| 3   | 2032 | 3,045 | 1,182 | 1,182 | 2,364 | 0.198 | 144%         |
| 3   | 2033 | 3,131 | 1,227 | 1,227 | 2,457 | 0.198 | 148%         |
| 3   | 2034 | 3,164 | 1,249 | 1,249 | 2,503 | 0.198 | 150%         |
| 3   | 2035 | 3,179 | 1,258 | 1,258 | 2,523 | 0.198 | 150%         |
| 3   | 2036 | 3,179 | 1,261 | 1,261 | 2,528 | 0.198 | 150%         |
| 3   | 2037 | 3,178 | 1,261 | 1,261 | 2,529 | 0.198 | 150%         |
| 3   | 2038 | 3,193 | 1,263 | 1,263 | 2,533 | 0.198 | 151%         |
| 4   | 2025 | 3,516 | 1,350 | 1,071 | 2,878 | 0.188 | 166%         |
| 4   | 2026 | 2,986 | 1,350 | 970   | 2,570 | 0.210 | 141%         |
| 4   | 2027 | 2,619 | 856   | 856   | 2,259 | 0.146 | 124%         |
| 4   | 2028 | 2,585 | 803   | 803   | 2,103 | 0.146 | 122%         |

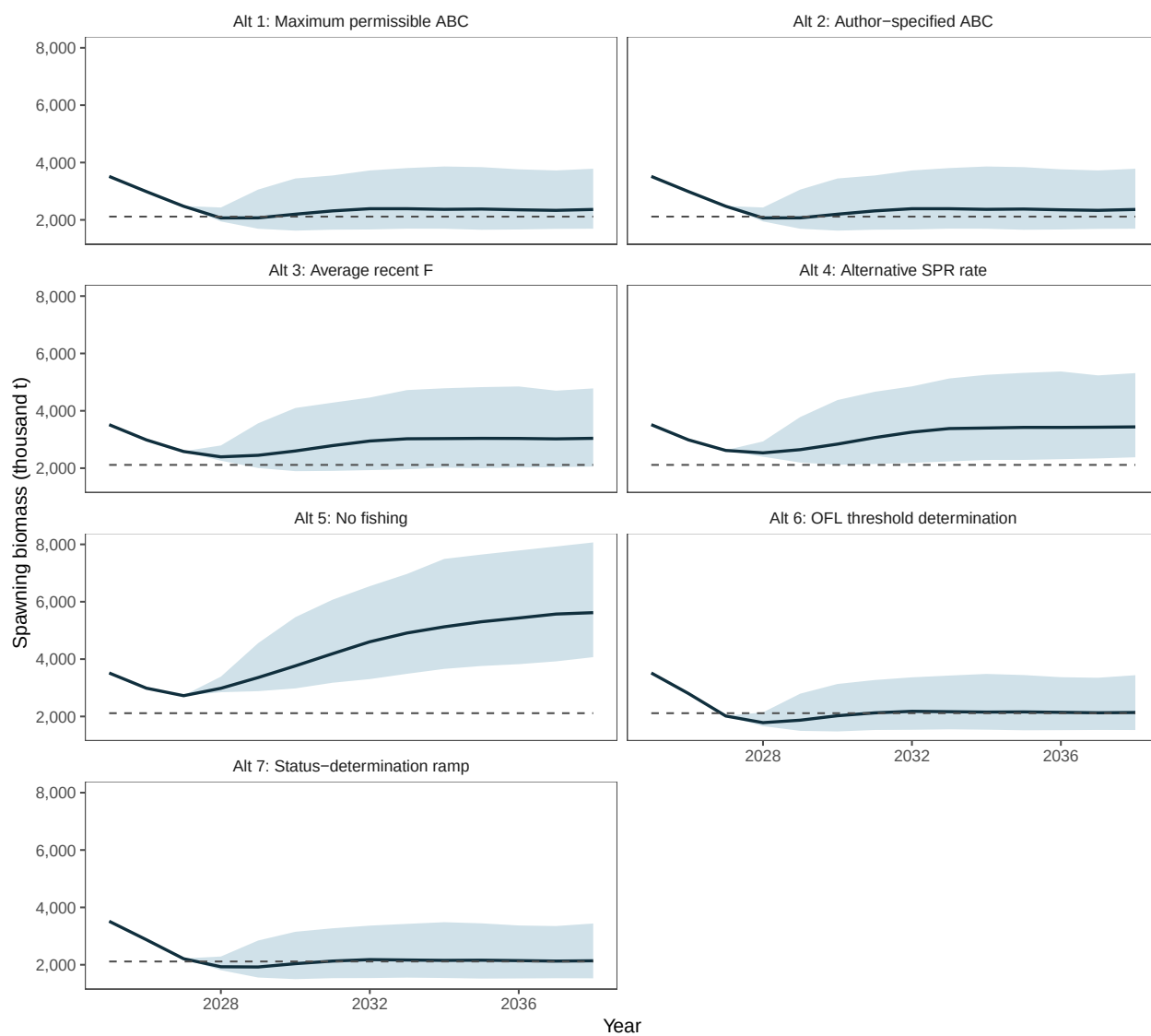


Figure 6: Projected spawning biomass under all seven Tier 3 alternatives. Lines are simulation medians, ribbons span the 5th to 95th percentiles, and dashed lines mark B35%.



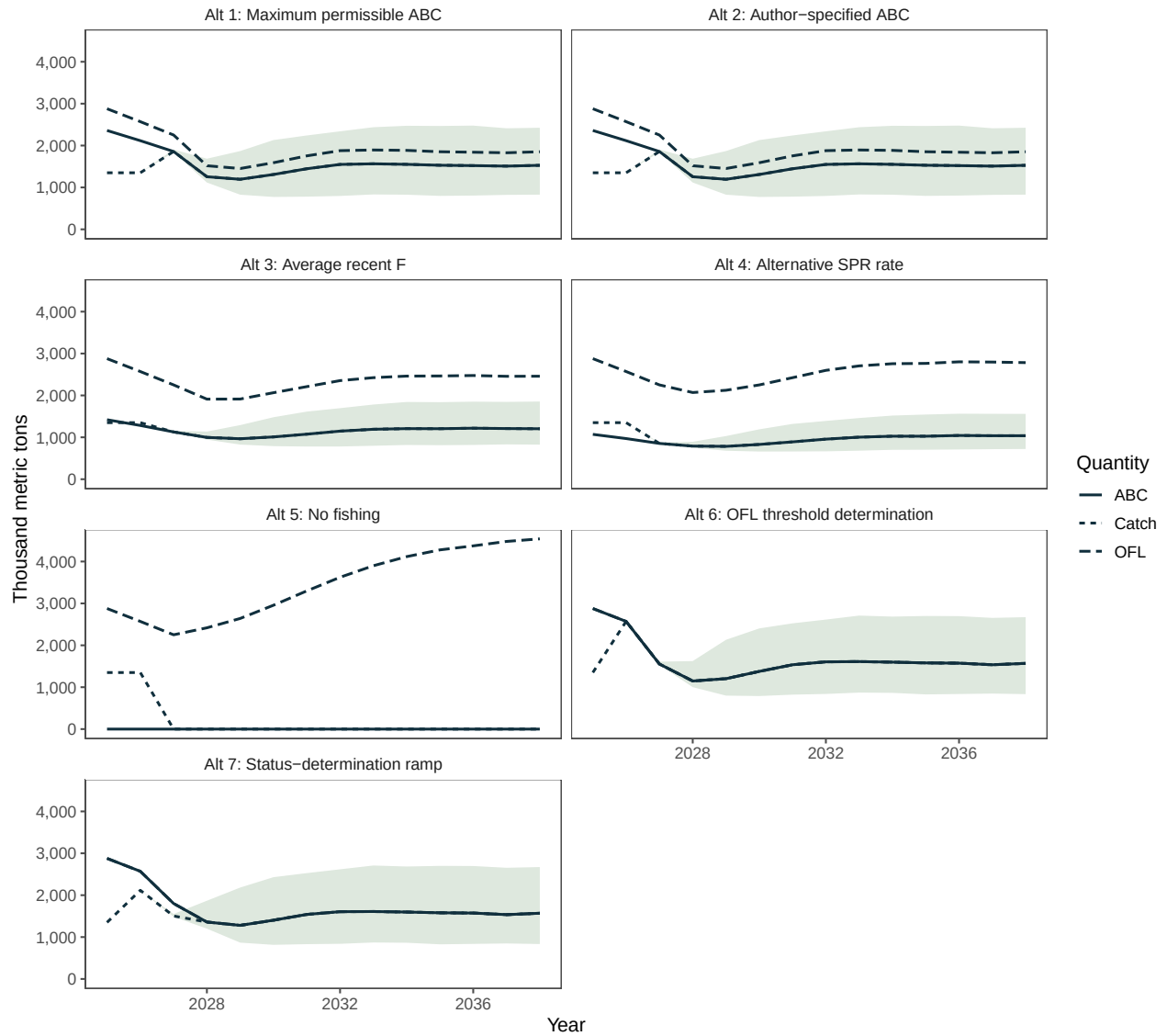


Figure 7: Projected catch, ABC, and OFL under all seven Tier 3 alternatives. Lines are simulation medians and catch ribbons span the 5th to 95th percentiles.

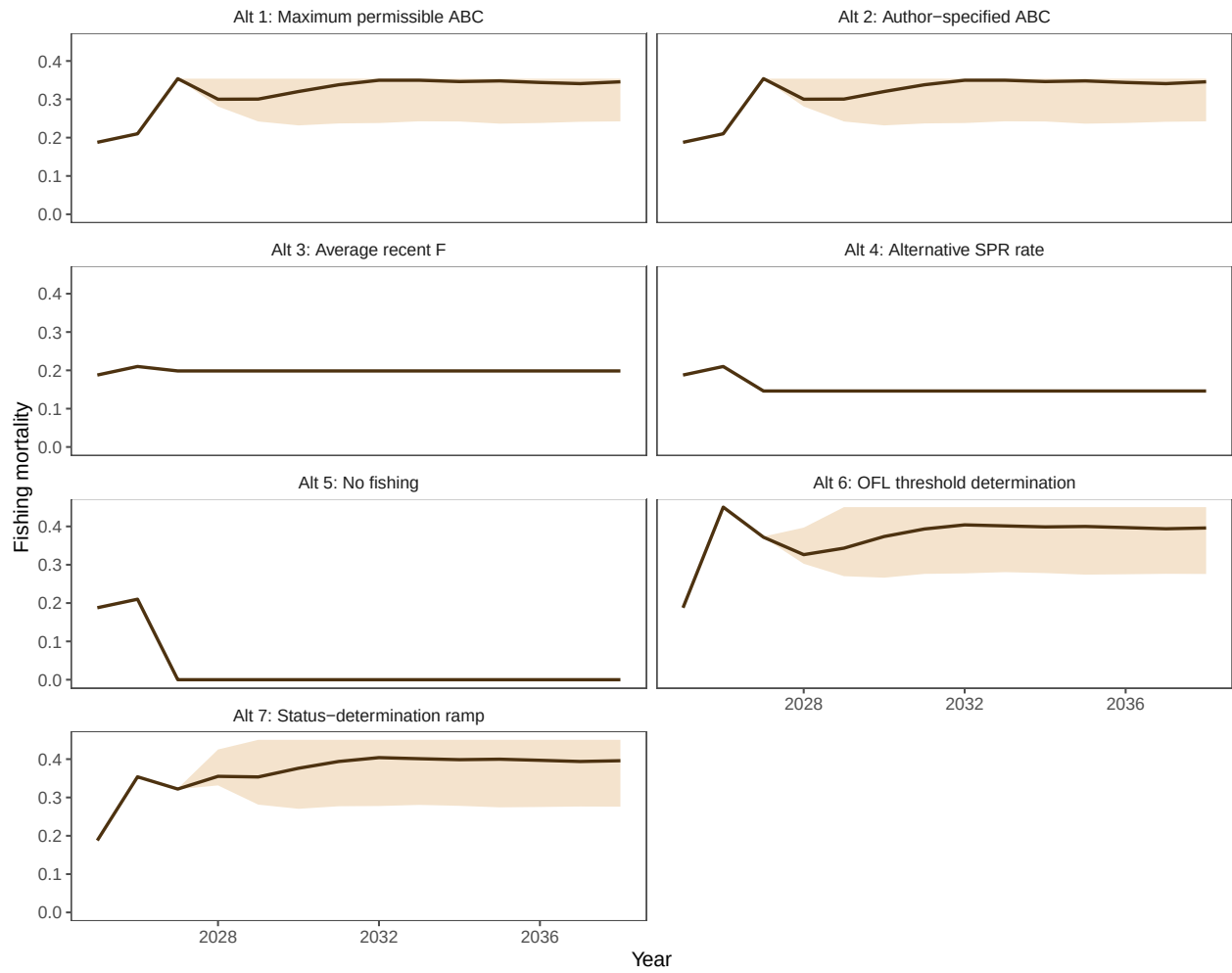


Figure 8: Projected fishing mortality under all seven Tier 3 alternatives. Lines are simulation medians and ribbons span the 5th to 95th percentiles.

## 7 Exploratory DSEM Recruitment Pathways

This section is the indicator-analysis pilot within a draft EBS pollock Ecosystem and Socioeconomic Profile (ESP), rather than an attempt to estimate the full pollock early-survival system. The BSAI Groundfish Plan Team recommended in 2021 that an EBS walleye pollock ESP be added to the list of future ESPs ([Bering Sea/Aleutian Islands Groundfish Plan Team 2021](#)). The analysis follows the sequence recommended in the AFSC causal-modeling guidance: define the outcome, build a literature-based mechanism table and generic DAG, select a small set of scientifically defensible graphs, retain a no-edge baseline, and compare all graphs on the same data ([Monnahan and Shotwell 2026](#)). The mechanism table and DAG draw on the January 2025 pollock workshop, the age-0 pollock working-group material, and the Tiger Team data-availability synthesis ([Monnahan et al. 2026](#); [Siddon et al. 2026](#); [Alaska Fisheries Science Center Age-0 Pollock Causal-Modeling Working Group 2026](#); [EBS Pollock Tiger Team 2026](#)) and the [EBS pollock Early Survival Pathways synthesis](#) ([J. N. Ianelli 2026](#)).

The response is the Rceattle age-1 recruitment state. The two examples test late-summer SST in the cohort’s age-0 year and cold-pool extent in the following age-1 calendar year. They were selected because both have relatively long annual records and represent distinct, prespecified hypotheses. The analysis does not treat them as direct measurements of prey quality, energetic condition, cannibalism, or survival.

### 7.1 ESP development scope

The September product documents the justification, synthesis, generic DAG, mechanism table, and two illustrative fits. It does not substitute a long list of available covariates for a small set of causal hypotheses. The intended progression to November is summarized in Table 14; the full draft ESP structure is provided in Appendix A-3.

Table 14: Development scope for the EBS pollock ESP and its Rceattle-DSEM indicator analysis.

| Component                 | September draft                                                                                    | November target                                                                                                 |
|---------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Justification and outcome | Document the Plan Team request and define age-1 recruitment as the focal population response.      | Confirm the focal response and management use with the ESP team.                                                |
| Indicator synthesis       | Provide a life-history synthesis, generic DAG, mechanism table, and data-availability review.      | Update indicators through the ESP and Ecosystem Status Report data-collection process.                          |
| Candidate hypotheses      | Retain two transparent proxy examples: age-0 late-summer SST and cold-pool extent.                 | Fit two to five defensible candidate DAGs that represent proximal mediators where data permit.                  |
| Indicator analysis        | Compare the two single-edge examples and their combined diagnostic with the no-edge IID reference. | Add predictive, retrospective, timing, and observation-error checks to the candidate-DAG comparison.            |
| Management use            | Use results to illustrate the integrated Rceattle-DSEM workflow and identify data gaps.            | Determine whether validated indicators are suitable for operational recruitment modeling or risk-table context. |

### 7.2 Mechanism table and generic DAG

The reduced mechanism table distinguishes processes that can be tested with the current annual data from biologically important mediators that remain intermittent or unavailable (Table 15). Expected signs describe the direction from the listed driver toward age-1 recruitment, whether directly or through the stated mechanism.

Table 15: Reduced mechanism table for the illustrative age-0 to age-1 recruitment analysis. Coverage refers to the cohort-aligned data presently available to this working paper.

| Driver                         | Mechanism                                                                            | Expected sign | Current coverage                                                                                       | Role here                      |
|--------------------------------|--------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------|--------------------------------|
| Sea ice                        | Winter ice affects summer cold-pool extent, bloom conditions, and prey production.   | Indirect      | Annual physical index available; not fitted here                                                       | Upstream context               |
| Cold-pool extent               | A larger cold pool can separate juveniles from adult pollock and reduce cannibalism. | Positive      | 42 annual values                                                                                       | Illustrative proxy edge        |
| Age-0 late-summer SST          | Warmer water can increase metabolic demand and reduce energy stored before winter.   | Negative      | 41 annual values                                                                                       | Illustrative proxy edge        |
| Large copepods and euphausiids | Lipid-rich prey supports age-0 growth, energy storage, and overwinter survival.      | Positive      | Intermittent; 15 values in the current ESP cohort table                                                | Unfitted mediator              |
| Fall age-0 energy              | Higher energy density and total energy increase overwinter survival to age 1.        | Positive      | Absent from the current annual cohort table                                                            | Priority mediator              |
| Adult-pollock overlap          | Greater adult-juvenile overlap increases cannibalism exposure.                       | Negative      | A separate range-overlap product exists, but its survey timing is contemporaneous with the age-1 state | Deferred from the main example |

The generic DAG is adapted from the age-0 pollock working-group graph and retains both bottom-up and top-down routes (Figure 9). It makes the unobserved or intermittent mediators visible instead of allowing the two available physical proxies to stand in for entire mechanisms. Solid arrows are conceptual. Dashed arrows are the only recruitment paths fitted in this proof of concept.

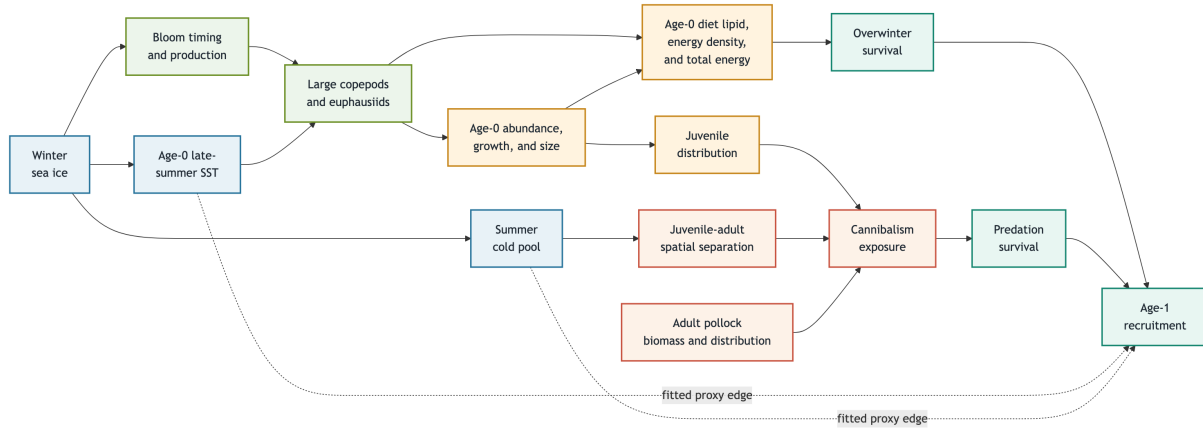


Figure 9: Generic directed acyclic graph for eastern Bering Sea pollock survival from age 0 to age 1. Solid arrows summarize hypothesized mechanisms. Dashed arrows are the two direct proxy paths fitted in the illustrative Rceattle-DSEM comparison.

The generic graph is intentionally broader than the statistical examples. It shows that a negative SST coefficient would be consistent with an energetic constraint, but would not identify prey quality, age-0 energy, or overwinter survival as the active mediator. Likewise, a cold-pool coefficient would be a proxy for spatial separation and cannibalism unless overlap and mortality were observed directly.

### 7.3 Two illustrative hypotheses

The proof of concept compares two single-edge hypotheses with a common reference:

- **Energetic-condition proxy:** higher age-0 late-summer SST is associated with lower age-1 recruitment because warmer conditions can increase metabolic demand and reduce fall energy reserves (Heintz et al. 2013; Siddon et al. 2013; Duffy-Anderson et al. 2016).
- **Spatial-separation proxy:** larger cold-pool extent is associated with higher age-1 recruitment because it can reduce juvenile-adult spatial overlap and cannibalism.

The age-0 SST value is carried forward to the DSEM row for that cohort's age-1 recruitment. The zero-lag path is therefore cohort-aligned rather than a claim that SST was measured in the age-1 year. In this

section “IID” refers to the no-recruitment-edge reference model. The combined model is retained as a limited diagnostic of whether the two proxy edges add information when fitted together; it is not a third mechanistic hypothesis.

All four specifications use the same observation-year universe, missingness pattern, standardization, and AR(1) processes for the observed covariates. Within this common data block, AIC compares relative fit after accounting for model complexity. The percentage reduction in residual recruitment variation compares each model’s estimated unexplained recruitment variance with the IID reference. It is an in-sample description, not a causal R-squared or a predictive-skill score.

## 7.4 Illustrative results

Table 16: Illustrative Rceattle-DSEM model comparison. Delta AIC and the percentage reduction in residual recruitment variation are calculated within the common observation block.

| Model                                | AIC      | Delta AIC | Residual<br>variation<br>reduced (%) |
|--------------------------------------|----------|-----------|--------------------------------------|
| Age-0 late-summer SST                | 4,144.56 | 0.00      | 10.5                                 |
| Cold pool + age-0<br>late-summer SST | 4,146.19 | 1.63      | 12.0                                 |
| IID recruitment<br>deviations        | 4,146.81 | 2.25      | 0.0                                  |
| Cold pool                            | 4,148.66 | 4.11      | 0.3                                  |

All four fits reached OK status, had positive-definite Hessians, and had maximum absolute gradients no larger than  $9.4\text{e-}05$ . The age-0 late-summer SST model had the lowest AIC (Table 16; Figure 10) and reduced estimated residual recruitment variation by 10.5% relative to IID. The cold-pool-only model reduced residual variation by 0.3% and had higher AIC than IID. The combined model reduced residual variation by 12.0%, but its additional parameter left its AIC above the SST-only model.

Table 17: Standardized proxy paths to age-1 recruitment. Intervals are the estimate plus or minus 1.96 standard errors.

| Covariate                         | Estimate | Standard error | Lower 95% | Upper 95% |
|-----------------------------------|----------|----------------|-----------|-----------|
| Cold pool                         |          |                |           |           |
| Cold pool                         | 0.041    | 0.107          | -0.169    | 0.251     |
| Age-0 late-summer SST             |          |                |           |           |
| Age-0 late-summer SST             | -0.222   | 0.104          | -0.426    | -0.017    |
| Cold pool + age-0 late-summer SST |          |                |           |           |
| Cold pool                         | -0.069   | 0.114          | -0.292    | 0.153     |
| Age-0 late-summer SST             | -0.254   | 0.118          | -0.485    | -0.024    |

The SST coefficient was negative and its interval excluded zero in both the single-edge and combined models. The cold-pool coefficient was small and its interval included zero in both models (Table 17; Figure 11). In plain language, the current annual data contain a modest in-sample association consistent with the warm-year

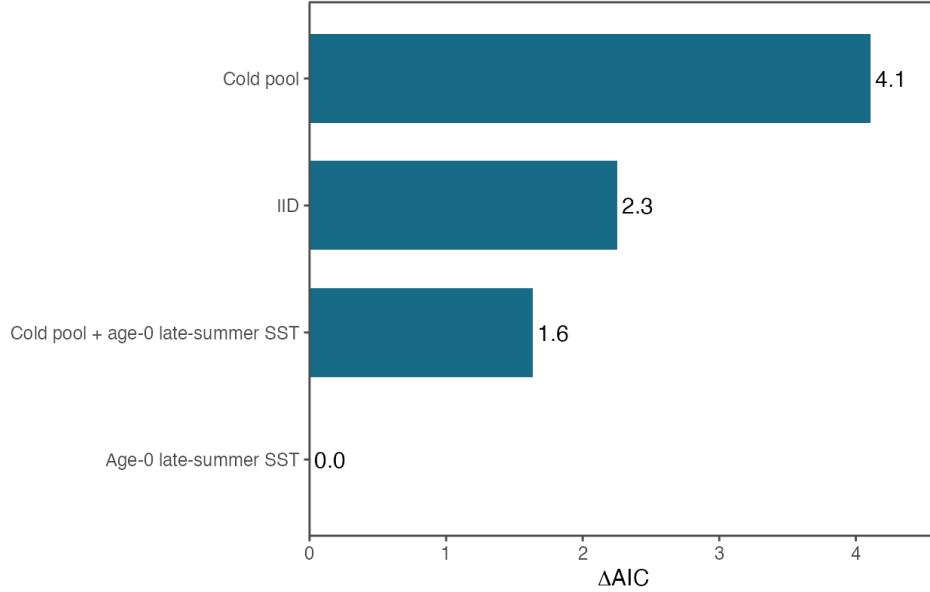


Figure 10: Akaike information criterion differences for the illustrative DSEM models. The age-0 late-summer SST model has the lowest AIC.

energetic-constraint hypothesis, while the direct cold-pool proxy edge received little support. Neither result establishes the intervening mechanism.

The missingness-preserving randomization check replaced both environmental series with independent standard-normal values using seed 20260812. None of the randomized recruitment paths improved on the randomized IID model: their delta AIC values ranged from 1.03 to 2.95, and every randomized path interval included zero. This is a basic check against an identical signal arising from arbitrary series with the same gaps; it does not replace out-of-sample validation.

## 7.5 Interpretation and next test

This proof of concept supports keeping the candidate set small. The SST result is useful as an illustration of the Rceattle-DSEM connection, but SST is an upstream proxy. The next scientifically stronger test is to insert a proximal mediator from the mechanism table: fall age-0 total energy or energy density, followed by a lipid-rich large-copepod/euphausiid index when enough cohort years are available (Eisner et al. 2020; Heintz et al. 2013; Siddon et al. 2013). That sequence would test whether the SST association weakens when the hypothesized energetic path is represented directly.

A top-down overlap example should return only after its timing and uncertainty are aligned with the age-0-to-age-1 survival interval. The available adult-pollock overlap product is informative, but its survey occurs after the start-of-year age-1 state and it is partly population-derived. Including it here would blur a causal example with a contemporaneous association.

Before either proxy is considered for operational recruitment inference, the paths should be tested with terminal-year or blocked prediction, retrospective stability, alternative cohort timing, and observation-error sensitivity. These DSEM results remain separate from the assessment configuration and projection results.

## 8 Summary

When Rceattle parameters and fishery selectivity are set to the values used in the ADMB model, Rceattle reproduces spawning biomass and predicted catch to numerical precision. This establishes agreement in

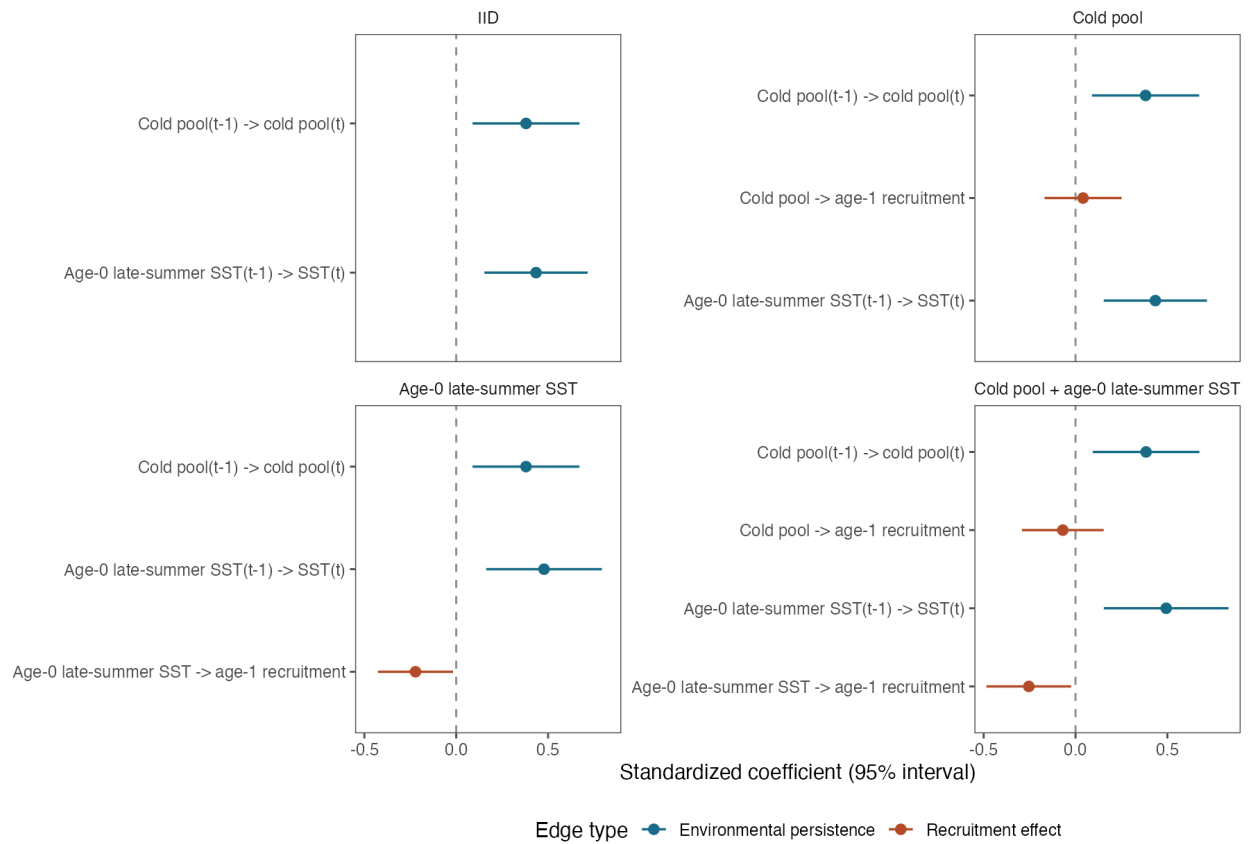


Figure 11: Standardized coefficients for the illustrative DSEM models. Points are estimates and horizontal lines are 95% intervals. Blue edges describe temporal persistence in observed covariates; orange edges are the two direct proxy paths to age-1 recruitment.

the tested population-dynamics calculations and parameter mapping. When Rceattle is fit to the data, the trajectories differ more, and the residual least-identified direction remains in the shared ATS/AVO annual selectivity deviations. The added ATS/AVO coefficient-increment penalty reduced the condition number below the convergence-warning threshold.

The diagnostic results identify a focused development path for the September 2026 Plan Team evaluation. The Rceattle fit reaches a small gradient, an OK convergence status, and a positive-definite Hessian. Composition OSA residuals remain underdispersed. Nine terminal-year-consistent peels converge successfully and produce positive retrospective bias in spawning biomass, total biomass, and recruitment, together with negative bias in fishing mortality. The historical 2D AR1 sensitivity did not converge and retains its research status. The 50-jitter diagnostic recovered the best solution 48 times and found a poorer mode twice; the age-3+ natural-mortality profile is shallow.

The scaled-back DSEM proof of concept compared two prespecified proxy edges with the no-edge IID reference. The age-0 late-summer SST model had the lowest AIC and reduced estimated residual recruitment variation by 10.5% relative to IID. The cold-pool edge received little support. This result illustrates the Rceattle-DSEM connection, but the annual data do not yet represent the prey, age-0 energy, overwinter-survival, and cannibalism mediators required to attribute the association to a causal mechanism.

Immediate work should retain the ADMB-to-Rceattle comparison as a repeatable numerical check, evaluate selectivity and survey influence on the retrospective pattern, and continue sensitivity checks for the shared ATS/AVO annual selectivity deviations. Given acceptance in other applications and the proven bridging aspects, we feel this approach should be adopted for operational management advice for November/December 2026. Future refinements will explore trade-offs in fitting data in alternative ways (e.g., using random effects approaches for selectivity variability, selecting alternatives to the robust-multinomial approach used in this example). The exploratory DSEM work provides a reproducible approach for adding cohort-aligned covariates, while remaining separate from the final assessment configuration and projection results.

## References

- Alaska Fisheries Science Center Age-0 Pollock Causal-Modeling Working Group. 2026. *Age-0 Pollock DAG*. Google Slides working document. <https://docs.google.com/presentation/d/1tninaCS9iSGh9aIXp7jGIcr1cE0owcr1E8yepBW2oYw/>.
- Alaska Fisheries Science Center Ecosystem and Socioeconomic Profile Coordination Team. 2026. *Blank ESP Tables and Figures*. Unpublished national ESP template.
- Bering Sea/Aleutian Islands Groundfish Plan Team. 2021. *Minutes of the Bering Sea/Aleutian Islands Groundfish Plan Team, September 22–23, 2021*. North Pacific Fishery Management Council meeting minutes. <https://meetings.npfmc.org/CommentReview/DownloadFile?p=f0e75d8e-4fbb-474a-a4aa-4c091cbc637b.pdf/&fileName=C6/%20September/%202021/%20BSAI/%20GPT/%20Minutes.pdf>.
- Duffy-Anderson, Janet T., Steven J. Barbeaux, Edward V. Farley, et al. 2016. “The Critical First Year of Life of Walleye Pollock (*Gadus chalcogrammus*) in the Eastern Bering Sea: Implications for Recruitment and Future Research.” *Deep-Sea Research Part II: Topical Studies in Oceanography* 134: 283–301. <https://doi.org/10.1016/j.dsr2.2015.02.001>.
- EBS Pollock Tiger Team. 2026. *EBS Pollock: Eco Framework*. Tiger Team Google Slides working document. <https://docs.google.com/presentation/d/1bSajTmPfWmiXqK6sHMOT8DZevklJ8ZzRrJEN2QdtIfw/>.
- Eisner, Lisa B., Ellen M. Yasumiishi, Alexander G. Andrews, and Cecilia A. O’Leary. 2020. “Large Copepods as Leading Indicators of Walleye Pollock Recruitment in the Southeastern Bering Sea: Sample-Based and Spatio-Temporal Model (VAST) Results.” *Fisheries Research* 232: 105720. <https://doi.org/10.1016/j.fishres.2020.105720>.



- Heintz, Ron A., Elizabeth C. Siddon, Edward V. Farley, and Jeffrey M. Napp. 2013. "Correlation Between Recruitment and Fall Condition of Age-0 Pollock (*Theragra chalcogramma*) from the Eastern Bering Sea Under Varying Climate Conditions." *Deep-Sea Research Part II: Topical Studies in Oceanography* 94: 150–56. <https://doi.org/10.1016/j.dsr2.2013.04.006>.
- Ianelli, James N. 2026. *Eastern Bering Sea Walleye Pollock Early Survival Pathways*. Quarto website. [https://afsc-assessments.github.io/ebswp\\_esp/](https://afsc-assessments.github.io/ebswp_esp/).
- Ianelli, Jim. 2026. *spmR: Standard Projection Model*. <http://afsc-assessments.github.io/spmR/>.
- Monnahan, Cole C., David McGowan, James N. Ianelli, Bridgett Ferriss, and Lauren A. Rogers, eds. 2026. *Proceedings of the Workshop on Walleye Pollock Biology and Ecology*. Draft Alaska Fisheries Science Center processed report.
- Monnahan, Cole C., and S. Kalei Shotwell. 2026. *Tentative Good Practices for AFSC Causal Modeling*. Living Alaska Fisheries Science Center working document. <https://docs.google.com/document/d/1ZyAQ-WtdqErHFxT567qQunvJI8WF-bi6yggpzkPHvUI/>.
- Schiano, Samantha, Sophie Breitbart, and Steve Saul. 2026. *Asar: Build NOAA Stock Assessment Report*. <https://github.com/nmfs-ost/asar>.
- Shotwell, S. Kalei, Jenny Bigman, and Russel Dame, eds. 2026. *Ecosystem and Socioeconomic Profile of the Arrowtooth Flounder Stock in the Gulf of Alaska*. Appendix 7A in the Gulf of Alaska arrowtooth flounder stock assessment.
- Siddon, Elizabeth C., Ron A. Heintz, and Franz J. Mueter. 2013. "Conceptual Model of Energy Allocation in Walleye Pollock (*Theragra chalcogramma*) from Age-0 to Age-1 in the Southeastern Bering Sea." *Deep-Sea Research Part II: Topical Studies in Oceanography* 94: 140–49. <https://doi.org/10.1016/j.dsr2.2012.12.007>.
- Siddon, Elizabeth C., S. Kalei Shotwell, and James N. Ianelli. 2026. *Age-0 Pollock DSEM Noteworthy: Eastern Bering Sea Walleye Pollock Early Survival Pathways*. Unpublished Alaska Fisheries Science Center working document.

## Appendices

### A-1 Reproducibility and analysis sequence

The working paper is rendered from saved model and diagnostic products so that building the document does not repeat long model fits. The commands below are run from the project directory, in the order shown when an earlier product is required by a later analysis.

1. The [input-preparation script](#) documents the transformations used to prepare the aligned Rceattle workbook; its precursor workbook is regenerated with Rscript "2024 EBS pollock setup data.R". The frozen workbook used for the reported fit is Data/EBS\_24\_pollock\_m23\_rceattle\_full\_1964-2024.xlsx. The same script documents the modified ADMB reference rebuild, run as `cd ADMB/m23_rceattle_full && admb pm && ./pm -nox -iprint 150` after the ADMB executable directory has been added to PATH.
2. The [two-stage fitting script](#) fits the reported Rceattle configuration and the 2D age-by-year AR1 sensitivity with Rscript "2024 EBS pollock.R". The [5.8.1 summary script](#) then generates the fitted-model summaries and standard-multinomial sensitivity using Rscript `R/finalize_rceattle_5_8_1_analysis.R`.
3. The [ADMB bridge runner](#) sets the Rceattle parameters and realized selectivity to the ADMB values and saves the trajectory, likelihood, and gradient comparisons with Rscript `R/run_canonical_pm_forward_bridge.R`.

4. The [OSA runner](#) generates the saved OSA residuals with Rscript `R/run_canonical_pm_diagnostics.R`. The [retrospective script](#) runs the nine terminal-year-consistent peels with Rscript `"2024 EBS pollock two-stage retrospectives.R"`.
5. The [5.8.1 self-test runner](#) generates 50 phased simulations with Rscript `R/run_rceattle_5_8_1_self_test.R`. The [restart runner](#) repeats the same simulations from the estimated solution with Rscript `R/run_rceattle_5_8_1_self_test_restart.R`, and the [validation summarizer](#) is run with Rscript `R/summarize_rceattle_5_8_1_validation.R`.
6. The [projection runner](#), which uses the [Rceattle-to-SPM input writer](#), generates the seven-scenario and fixed-catch projection products with Rscript `R/run_spmr_tier3_projection.R`.
7. The [DSEM script](#) runs the four observed-covariate models with Rscript `"2024 EBS pollock DSEM proof of concept.R"` and the missingness-preserving randomized controls with Rscript `"2024 EBS pollock DSEM proof of concept.R" --randomized`.
8. After these saved products are available, the HTML and PDF are generated together with `quarto render ebswp.qmd`.

R version 4.6.1 (2026-06-24)

Platform: aarch64-apple-darwin23

Running under: macOS Tahoe 26.6.2

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK version 3.12.0

locale:

[1] C.UTF-8/C.UTF-8/C.UTF-8/C.UTF-8/C.UTF-8

time zone: America/Los\_Angeles

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] tidyr\_1.3.2 tibble\_3.3.1 scales\_1.4.0 readr\_2.2.0  
 [5] Rceattle\_5.8.1 knitr\_1.51 gt\_1.3.0 ggthemes\_5.2.0  
 [9] ggplot2\_4.0.3 ggridges\_0.5.7 dplyr\_1.2.1 asar\_2.5.0.9000  
 [13] afscOSA\_0.0.1

loaded via namespace only:

[1] generics\_0.1.4 xml2\_1.6.0 stringi\_1.8.7 lattice\_0.22-9  
 [5] hms\_1.1.4 digest\_0.6.39 magrittr\_2.0.5 evaluate\_1.0.5  
 [9] grid\_4.6.1 RColorBrewer\_1.1-3 fastmap\_1.2.0 plyr\_1.8.9  
 [13] jsonlite\_2.0.0 Matrix\_1.7-5 tinytex\_0.60 purrr\_1.2.2  
 [17] viridisLite\_0.4.3 cli\_3.6.6 crayon\_1.5.3 rlang\_1.3.0  
 [21] litedown\_0.9 commonmark\_2.0.0 cowplot\_1.2.0 bit64\_4.8.2  
 [25] withr\_3.0.3 yaml\_2.3.12 otel\_0.2.0 parallel\_4.6.1  
 [29] tools\_4.6.1 reshape2\_1.4.5 tzdb\_0.5.0 vctrs\_0.7.3  
 [33] R6\_2.6.1 lifecycle\_1.0.5 stringr\_1.6.0 bit\_4.6.0  
 [37] fs\_2.1.0 vroom\_1.7.1 pkgconfig\_2.0.3 pillar\_1.11.1  
 [41] gtable\_0.3.6 glue\_1.8.1 Rcpp\_1.1.2 xfun\_0.60  
 [45] tidyselect\_1.2.1 farver\_2.1.2 htmltools\_0.5.9 labeling\_0.4.3  
 [49] rmarkdown\_2.31 TMB\_1.9.21 compiler\_4.6.1 S7\_0.2.2  
 [53] markdown\_2.0

Table 18: Files used or referenced by this working paper.

| Role                                        | Location                                | Available | Size        |
|---------------------------------------------|-----------------------------------------|-----------|-------------|
| Prepared model-input workbook               | Local workbook                          | TRUE      | 135.1 KB    |
| BTS covariance input                        | Local data file                         | TRUE      | 12.7 KB     |
| Saved fitted Rceattle objects               | Local R serialized-data (RDS) file      | TRUE      | 88014.2 KB  |
| OSA residuals                               | Local RDS file                          | TRUE      | 91.2 KB     |
| OSA diagnostics                             | Local RDS file                          | TRUE      | 1.1 KB      |
| Nine-peel Rceattle retrospective            | Local RDS file                          | TRUE      | 368646.6 KB |
| Nine-peel Mohn's rho summary                | Local CSV file                          | TRUE      | 0.2 KB      |
| Nine-peel retrospective script              | Local R script                          | TRUE      | 9.7 KB      |
| ADMB-to-Rceattle comparison output          | Local RDS file                          | TRUE      | 650.8 KB    |
| Rceattle 5.8.1 self-test summary            | Local CSV file                          | TRUE      | 0.5 KB      |
| Rceattle 5.8.1 truncation diagnostics       | Local CSV file                          | TRUE      | 0.4 KB      |
| Rceattle 5.8.1 restart comparison           | Local CSV file                          | TRUE      | 3.8 KB      |
| Rceattle 5.8.1 jitter summary               | Local CSV file                          | TRUE      | 2.3 KB      |
| Age-3+ natural-mortality profile            | Local CSV file                          | TRUE      | 0.3 KB      |
| Age-3+ natural-mortality profile components | Local CSV file                          | TRUE      | 10.5 KB     |
| Age-3+ natural-mortality fleet components   | Local CSV file                          | TRUE      | 48.2 KB     |
| Natural-mortality profile summarizer        | Local R script                          | TRUE      | 3.8 KB      |
| ADMB-to-Rceattle comparison summary         | Local comma-separated-values (CSV) file | TRUE      | 1.0 KB      |
| Rceattle fitting script                     | Local R script                          | TRUE      | 15.7 KB     |
| Diagnostics script                          | Local R script                          | TRUE      | 4.9 KB      |
| Next-assessment update script               | Local R script                          | TRUE      | 2.6 KB      |
| ADMB comparison script                      | Local R script                          | TRUE      | 10.1 KB     |
| Rceattle-to-SPM input writer                | Local R script                          | TRUE      | 8.7 KB      |
| spmR Tier 3 projection runner               | Local R script                          | TRUE      | 1.1 KB      |
| SPM projection manifest                     | Local CSV file                          | TRUE      | 0.5 KB      |
| Exploratory DSEM script                     | Local R script                          | TRUE      | 16.8 KB     |
| DSEM model comparison                       | Local CSV file                          | TRUE      | 0.5 KB      |
| DSEM recruitment-path estimates             | Local CSV file                          | TRUE      | 3.1 KB      |
| Randomized DSEM control                     | Local CSV file                          | TRUE      | 0.5 KB      |

## A-2 Rceattle-to-spmR projection bridge

The projection bridge is a deterministic file adapter rather than a second assessment fit. The `input writer` reads the saved `nonparametric_pm` Rceattle object, extracts the population state, biological schedules, fishery schedule, and historical series needed by the Standard Projection Model (SPM), and writes them in the SPM `pm.prj` format. It also writes the projection controls in `spm.dat`. The `projection runner` passes those files to `spmR::runSPM(..., engine = "admb")`, which executes the validated SPM binary and returns the simulation-level output used by the report.

The adapter preserves the terminal assessment state and makes each conversion explicit (Table 19). Numbers at age, natural mortality, and the weight schedules are passed on their Rceattle scales. With abundance in millions of fish and weight in kilograms, their product has units of thousand metric tons. Maturity is divided by its maximum, matching the SPM input convention used for this stock. The spawning-month encodings differ by one because Rceattle records elapsed months and SPM uses a one-based calendar month; both represent survival through 0.25 of the year.

Table 19: Mapping from the saved Rceattle fit to the SPM species input file.

| SPM field                  | Rceattle source                    | Transformation and years                             |
|----------------------------|------------------------------------|------------------------------------------------------|
| Average recent F           | quantities: F_spp                  | Arithmetic mean for 2020–2024; used by Alternative 3 |
| Natural mortality at age   | quantities: M_at_age               | Terminal-year schedule (2024)                        |
| Maturity at age            | data_list: maturity                | Divided by its maximum to match the SPM convention   |
| Spawning weight at age     | data_list: weight and ssb_wt_index | Terminal-year spawning-weight schedule               |
| Fishery weight at age      | data_list: weight, fishery index   | Terminal-year fishery-weight schedule                |
| Fishery selectivity at age | quantities: sel_at_age for Fishery | Arithmetic mean at each age for 2020–2024            |
| Terminal numbers at age    | quantities: N_at_age               | Terminal assessment state (2024)                     |
| Recruitment history        | quantities: R                      | Annual age-1 recruitment for 1978–2024               |
| Spawning-biomass history   | quantities: ssb                    | Annual spawning biomass for 1977–2023                |
| Spawning month             | data_list: spawn_month = 3         | Written as SPM Spawnmo = 4; both equal 0.25 year     |

Annual Rceattle fishery selectivity is already normalized to a mean of one across ages. The projection schedule is calculated as the arithmetic mean at each age across 2020–2024; averaging these normalized annual curves retains a mean of one. This five-year schedule replaces the former terminal-year-only curve. The writer saves the five contributing curves in `fishery_selectivity_recent_years.csv`, the applied average in `age_schedules.csv`, and the rule and exact years in `lineage.csv`. Thus the projection input can be reconstructed without reading values back from the formatted `pm.prj` file.

The `spm.dat` file starts projections in 2025, runs 14 projection years and 1,000 simulations, and requests all seven Tier 3 alternatives. Catch is fixed at 1,350 thousand t in 2025 and 2026 before the alternatives diverge. A separate call uses Alternative 2 with catch fixed at 1,300 thousand t for 2025–2032. The writer copies the validated SPM executable and `tacpar.dat` control file into each run directory. `manifest.csv` records the

role, presence, and MD5 checksum of every required input, while `lineage.csv` records the Rceattle source checksum, terminal year, Rceattle and spmR versions, selectivity rule, and averaging years.

The saved products passed the bridge checks in Table 20. These checks establish file and field consistency; they do not replace review of the harvest assumptions or the biological appropriateness of a five-year selectivity average.

Table 20: Automated consistency checks for the Rceattle-to-spmR projection bridge.

| Check                              | Result      | Evidence                                   |
|------------------------------------|-------------|--------------------------------------------|
| Required projection files          | <b>Pass</b> | 6 of 6 manifest targets present            |
| Recent selectivity years           | <b>Pass</b> | 2020, 2021, 2022, 2023, 2024               |
| Applied five-year selectivity mean | <b>Pass</b> | maximum absolute difference = 3.996803e-15 |
| Projection alternatives            | <b>Pass</b> | 1, 2, 3, 4, 5, 6, 7                        |
| Projection start year              | <b>Pass</b> | 2025                                       |

### A-3 Draft EBS pollock ESP development framework

The most effective way to integrate the ecosystem work into this working paper is to keep the concise DSEM pilot in Section 7 and develop the full ESP as an appendix. This preserves the assessment narrative while giving the ESP its own standard structure, evidence trail, and update cycle. The scaffold below adapts the national ESP tables and the 2026 Gulf of Alaska arrowtooth flounder ESP ([Alaska Fisheries Science Center Ecosystem and Socioeconomic Profile Coordination Team 2026](#); [Shotwell et al. 2026](#)). It is a September draft: the indicator synthesis and proposed causal structure are present, while the final indicator updates, expanded candidate-DAG comparison, and management summary are November deliverables.

Table 21: Proposed EBS pollock ESP appendix structure and status in this working paper.

| ESP component                  | EBS pollock content                                                                                                                                        | Current status                                                                              |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Executive summary              | ABC/TAC relevance, predictive and contextual indicators, principal data gaps.                                                                              | Write after the November indicator analysis.                                                |
| Introduction and justification | 2021 BSAI Plan Team request; focal response is age-1 recruitment and its early-survival mechanisms.                                                        | Justification documented; team review pending.                                              |
| Data                           | Assessment states, physical indices, plankton and prey indices, age-0 condition, predation/overlap products, and selected socioeconomic series.            | Inventory started; several proximal biological series remain short or intermittent.         |
| Indicator synthesis            | Life-history table, conceptual diagram, generic DAG, and mechanism table developed from the workshop and working-group material.                           | Draft synthesis presented here and in Section 7.                                            |
| Proposed indicators            | A small set selected by mechanism, timing, record length, uncertainty, and availability rather than correlation alone.                                     | SST and cold pool illustrate the workflow; age-0 energy and lipid-rich prey are priorities. |
| Indicator assessment           | Current condition, trends, uncertainty, and interpretation for ecosystem and socioeconomic indicators.                                                     | Complete after current-year ESP and ESR data collection.                                    |
| Indicator analysis             | No-edge reference, candidate DAGs, common-data AIC, path estimates, residual-variance change, predictive checks, and scientific sign checks.               | Two proxy examples completed; expanded candidate-DAG analysis deferred to November.         |
| Data gaps and priorities       | Fall age-0 energy, prey lipid/abundance, bloom-prey match timing, juvenile-adult overlap, and consistent observation-error information.                    | Priorities identified; feasible annual products require team confirmation.                  |
| Conclusion and management use  | Separate predictive indicators from contextual indicators and state whether each is suitable for recruitment modeling, the risk table, or monitoring only. | Complete after validation and team review.                                                  |

The draft life-history synthesis in Table 22 condenses the standard ESP life-history table to the stages and information needed for the recruitment question. A later stand-alone ESP can expand each stage into the national-template fields for habitat and distribution, phenology, growth, energetics, diet, and predators or competitors.

Table 22: Draft life-history synthesis for EBS pollock recruitment, condensed from the national ESP template.

| Stage                     | Process emphasized                                                                                   | Available evidence                                                                  | Priority gap                                                                         |
|---------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Spawning and eggs         | Adult condition, spawn timing and location, egg development, and transport.                          | Adult condition and physical indices are regularly available.                       | Cohort-specific egg production and transport exposure.                               |
| Larvae                    | Bloom timing, prey match, transport, temperature-dependent growth, and predation.                    | Physical forcing is available; biological observations are less consistent.         | Annual bloom-prey match and larval survival indices.                                 |
| Young of year             | Prey quantity and lipid quality, growth, distribution, and accumulation of fall energy.              | SST is long; prey and condition series are shorter or intermittent.                 | Consistent fall age-0 abundance, diet lipid, energy density, and total energy.       |
| Overwinter age 0 to age 1 | Energy reserves, metabolic demand, winter severity, and predation determine survival.                | Physical proxies and modeled recruitment are available.                             | Direct annual overwinter-survival observations and uncertainty.                      |
| Juvenile and pre-recruit  | Spatial overlap with adults and other predators affects cannibalism and natural mortality.           | Adult abundance is long; overlap and mortality products require timing review.      | Cohort-aligned juvenile-adult overlap and predation mortality.                       |
| Adult                     | Spawner abundance, condition, distribution, and cannibalism affect production and juvenile survival. | Assessment abundance, maturity, distribution, and condition products are available. | Separate maternal and density-dependent pathways from shared assessment information. |

The generic DAG in Section 7 serves as the initial full conceptual graph. The fitted SST and cold-pool graphs are deliberately simplified descendants of that graph, not a search across every available series. For November, each candidate graph should be traceable to a row in the mechanism table and should use the same analysis data block as the IID no-recruitment-edge reference. The final ESP should then report the causal statement, lag, expected sign, data limitations, path estimate, uncertainty, change in unexplained recruitment variation, and predictive check for every retained recruitment edge.

Fishery performance and socioeconomic indicators can be transferred from the SAFE after the biological indicator set is stable. They should occupy the standard socioeconomic synthesis and monitoring subsections rather than be inserted as recruitment causes unless a separate, defensible causal hypothesis requires them. This division keeps the integrated Rceattle-DSEM analysis focused on the population response while allowing the ESP appendix to inform both biological risk and TAC considerations.

## A-4 Model fitting details

The fishery selectivity likelihood has multiple modes, so the reported Rceattle configuration used data-based fishery-selectivity starting values and two fitting stages. The first stage disabled the annual selectivity deviations to establish the selectivity and abundance scale. The second stage restored the deviations and used the first-stage solution as its initial values. This sequence produced a small maximum gradient and OK status. With the direct ATS/AVO coefficient-increment penalty, the Hessian condition number was approximately 489,000; its weakest direction remained in the shared ATS/AVO annual selectivity deviations (`sel_coff_dev`) rather than fishery/CPUE.

The complete fitting sequence, including the empirical starting values and the two estimation stages, is in the [two-stage fitting script](#) and is run with `Rscript "2024 EBS pollock.R"`. The nine-peel implementation is in the [retrospective script](#) and is run with `Rscript "2024 EBS pollock two-stage retrospectives.R"`.

The retrospective peels repeat this sequence after recalculating the data-based selectivity starting values for each terminal year. Because `sel_coff_dev` contains annual random-walk increments at age, each peel fixes the terminal-year fishery and mirrored catch-per-unit-effort increments at zero. This holds their terminal selectivity-at-age curves at the preceding-year curves.

## A-5 Jitter and natural-mortality profile

The [extended-diagnostics runner](#) generates both the 50 jitter fits and the age-3-and-older natural-mortality profile with `Rscript R/run_rceattle_5_8_1_extended_diagnostics.R`. The [profile-component summarizer](#) extracts the saved `jnl_comp` matrices without refitting the model and is run with `Rscript R/summarize_rceattle_5_8_1_m_profile.R`.

The 50 phased jitter runs test sensitivity to dispersed starting values. Forty-eight returned to the best objective within numerical precision. Two received WARN status and reached an inferior mode with an objective increase of approximately 609.5. This reproduces the pre-existing multimodality and shows why convergence of one run is not sufficient evidence that the global mode was found.

The age-3-and-older natural-mortality profile was evaluated from 0.15 to 0.50. The grid minimum occurred at 0.20, but the objective changed by only 3.82 across the entire grid. The profile is therefore shallow and does not support precise estimation of natural mortality from these data alone.

The saved profile fits retain Rceattle's full negative-log-likelihood (NLL) component matrix. The component entries sum to the reported objective at every grid point to within numerical precision. In Figure 12, each component is centered on its value at the grid minimum of 0.20. Positive changes indicate that a component favors the lower grid-minimum value over the plotted value; negative changes indicate that the component favors the plotted value. The total panel is the sum of the data and penalty panels. The M-prior contribution is zero in this configuration.

Table 23: Disposition of the Rceattle development components.

| Item     | Development component                   | Review result                                                                                                                                   | Interpretation                                                                                                                   |
|----------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | Prepared Rceattle model-input workbook  | Used the delivered workbook directly.                                                                                                           | Input files are available and reproducible from stored files.                                                                    |
| <b>2</b> | Rceattle fit and 2D AR1 sensitivity     | The Rceattle fit completed; 2D AR1 received a FAIL convergence status.                                                                          | The NonParametricPM configuration is the relevant Rceattle comparison; 2D AR1 is diagnostic only.                                |
| <b>3</b> | Convergence and model diagnostics       | Structured convergence checks, OSA diagnostics, nine retrospective peels, 50 self-tests, 50 jitters, and a natural-mortality profile are saved. | The ATS/AVO coefficient-increment penalty resolves the prior Hessian-conditioning warning; two inferior jitter solutions remain. |
| <b>4</b> | Next-assessment data update workflow    | Update script review is complete; execution awaits complete new-year observations.                                                              | Useful as a template only after complete new-year observations are inserted.                                                     |
| <b>5</b> | ADMB-to-Rceattle comparison diagnostics | Setting Rceattle to the ADMB values reproduces ADMB to rounding-scale precision.                                                                | This is the strongest evidence that the ADMB comparison is correct.                                                              |

Table 24: Rceattle 5.8.1 phased-jitter results for the fitted configuration. Delta objective is measured from the best of 50 runs.

| Status | Runs | Minimum delta objective | Maximum delta objective |
|--------|------|-------------------------|-------------------------|
| OK     | 48   | 0.000                   | 0.000                   |
| WARN   | 2    | 609.548                 | 609.548                 |



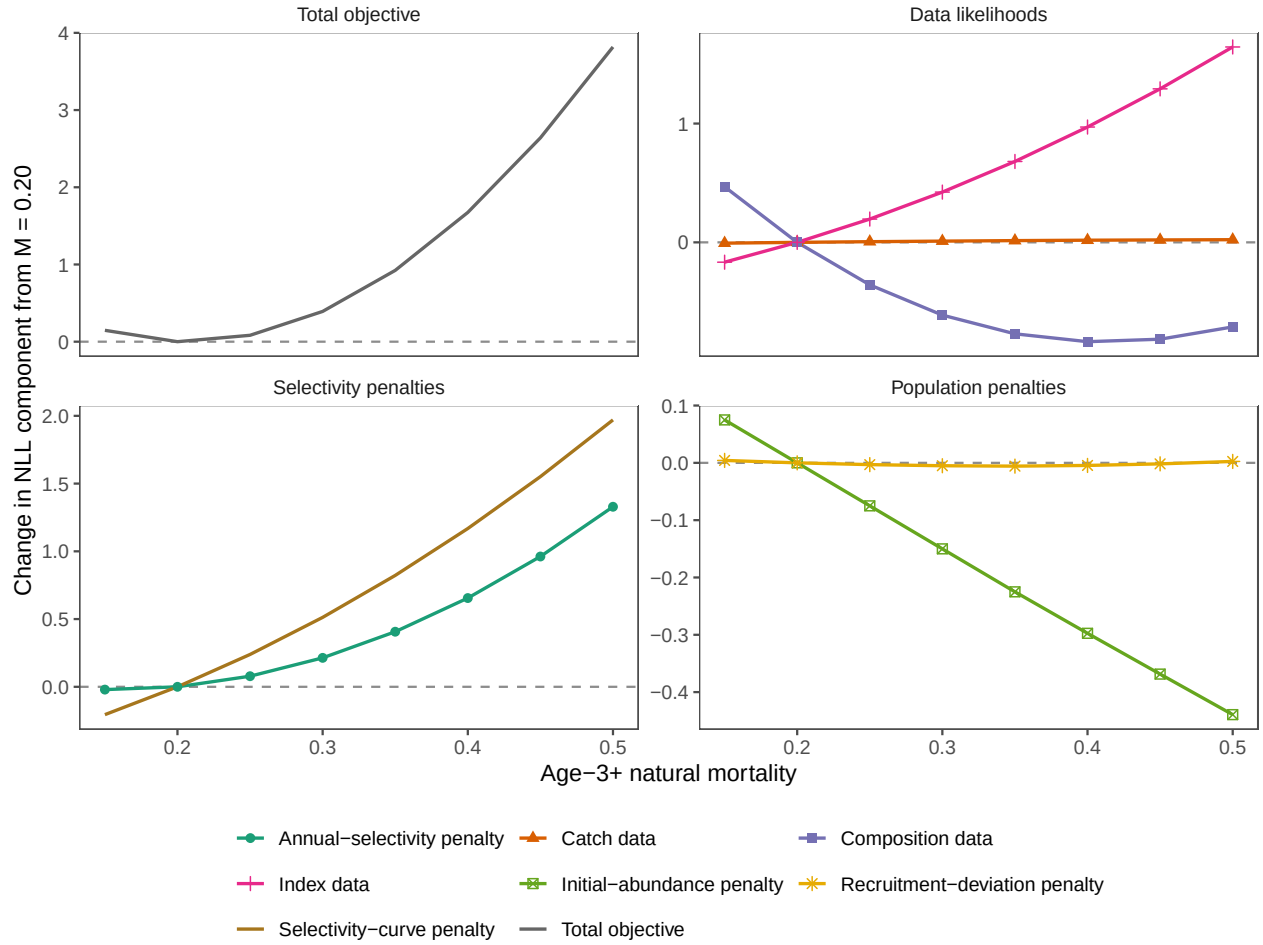


Figure 12: Component decomposition of the age-3-and-older natural-mortality profile. Each curve is the change in that NLL component relative to its value at  $M = 0.20$ . Positive values favor  $M = 0.20$  over the plotted value; negative values favor the plotted value. Components in the three lower panels sum to the total objective change in the upper-left panel.

The increase in total NLL from ( $M=0.20$ ) to ( $M=0.50$ ) was 3.82. The material fleet-specific contributions to that change are shown in Table 25. The strongest single counterweight was the BTS age-composition likelihood, which improved by 5.27 NLL units as  $M$  increased. Lower  $M$  was favored mainly by the ATS age-composition likelihood, the BTS index likelihood, and the selectivity penalties. Because ATS and AVO share a selectivity index in this configuration, the shared ATS/AVO penalties are stored in the AVO-labeled column of `jnll_comp`.

Table 25: Material fleet-specific contributions to the total NLL change from  $M = 0.20$  to  $M = 0.50$ . The direction column identifies the  $M$  value favored by each term; positive contributions favor lower  $M$  and negative contributions favor higher  $M$ . Terms with absolute contributions below 0.25 NLL units are omitted.

| Term                                      | Direction<br>favored | NLL<br>contribution |
|-------------------------------------------|----------------------|---------------------|
| ATS age-composition likelihood            | <b>Lower M</b>       | 3.40                |
| Shared ATS/AVO annual-selectivity penalty | <b>Lower M</b>       | 2.59                |
| BTS index likelihood                      | <b>Lower M</b>       | 2.52                |
| Shared ATS/AVO selectivity-curve penalty  | <b>Lower M</b>       | 1.32                |
| Fishery age-composition likelihood        | <b>Lower M</b>       | 1.16                |
| Fishery selectivity-curve penalty         | <b>Lower M</b>       | 0.65                |
| ATS index likelihood                      | <b>Higher M</b>      | -0.26               |
| Population initial-abundance penalty      | <b>Higher M</b>      | -0.44               |
| AVO index likelihood                      | <b>Higher M</b>      | -0.52               |
| BTS annual-selectivity penalty            | <b>Higher M</b>      | -1.35               |
| BTS age-composition likelihood            | <b>Higher M</b>      | -5.27               |

This decomposition shows that the low profile minimum is a trade-off among specific survey, composition, and selectivity terms rather than a uniform signal for lower natural mortality. It also explains why this result can differ from natural-mortality analyses that use different composition likelihoods, survey weights, or selectivity penalties. A direct comparison with those analyses should match these components and their weights before interpreting the difference as biological evidence about  $M$ .

## A-6 Composition-likelihood sensitivity

The [5.8.1 summary script](#) fits the standard-multinomial sensitivity and writes the summaries used here when run with `Rscript R/finalize_rceattle_5_8_1_analysis.R`.

The Rceattle configuration uses the supported `MultinomialAFSC` composition likelihood with a robustness constant of 0.001 added to observed and expected proportions. A standard multinomial fit without this constant was retained as an observation-model sensitivity. Both fits reached small gradients and received `OK` status. The final configuration's residual least-identified direction remained in the shared ATS/AVO annual selectivity deviations.

Table 26: Composition-likelihood sensitivity for the Rceattle 5.8.1 configuration. Raw objectives are shown for run provenance but are not compared because likelihood constants and conventions differ.

| Model                | Composition likelihood | Objective | Maximum gradient      | Fit status |
|----------------------|------------------------|-----------|-----------------------|------------|
| Final configuration  | AFSC multinomial       | 716.948   | $4.14 \times 10^{-5}$ | OK         |
| Standard multinomial | Standard multinomial   | 2,843.416 | $2.80 \times 10^{-4}$ | OK         |

Relative to the `MultinomialAFSC` configuration, the standard multinomial sensitivity changed spawning biomass by 1.79%, recruitment by 1.05%, and total biomass by 11.09% on average in absolute percentage terms.

## A-7 2D age-by-year AR1 selectivity sensitivity

The penalized-effects initialization and subsequent Laplace fit are in the [two-stage fitting script](#) and are run with `Rscript "2024 EBS pollock.R"`. A separate [fixed-scale follow-up](#) can be run with `Rscript "2024 EBS pollock 2D AR1 fixed sd test.R"`; it is a development check and is not the sensitivity summarized below.

The 2D AR1 sensitivity applied an age-by-year field to the shared fishery and catch-per-unit-effort selectivity block over 1964–2024. The age-year deviations were first fit as penalized effects to initialize the field and then integrated as random effects with a Laplace approximation. The initialized Laplace fit failed the gradient, Hessian, and bound checks.

Table 27: Diagnostic 2D AR1 fishery-selectivity correlation and scale estimates.

| Quantity                    | Rceattle parameter | Penalized effects | Random-effects fit |
|-----------------------------|--------------------|-------------------|--------------------|
| Year AR1 correlation        | Sel_curve_pen1     | 0.584             | 0.947              |
| Age AR1 correlation         | Sel_curve_pen2     | 0.120             | 0.098              |
| Selectivity deviation scale | sel_dev_log_sd     | 0.500             | 0.379              |

Correlations are on the -1 to 1 scale; the deviation scale is a positive standard deviation. The fit failed convergence checks.

The failed random-effects fit had a maximum marginal gradient of 19.7 on `rec_pars`, a non-invertible Hessian, 61 `log_F` parameters at their configured bounds, and `FAIL` convergence status. These estimates remain diagnostic.

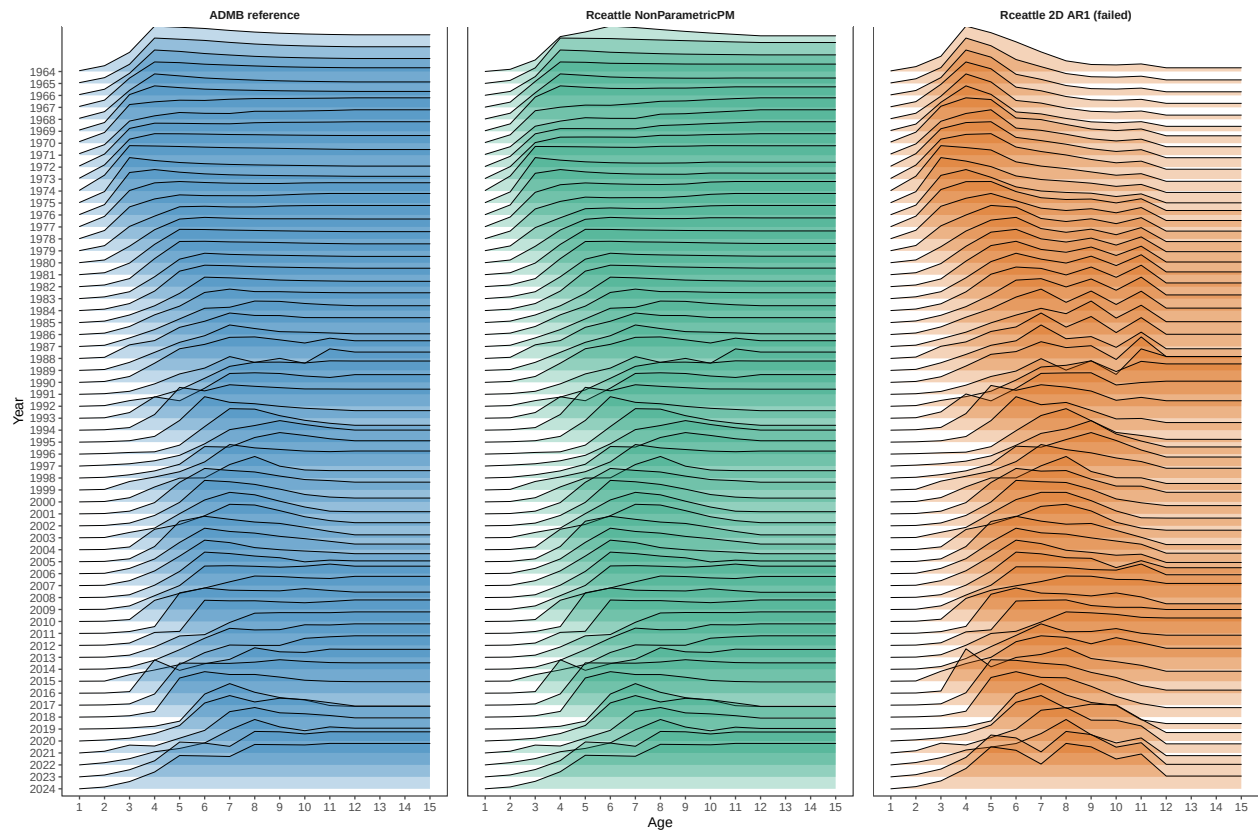


Figure 13: Development comparison of relative fishery selectivity for ADMB, the Rceattle configuration, and the failed 2D age-by-year AR1 sensitivity.

Table 28: Acronyms used in this working paper.

| Acronym | Definition                                 |
|---------|--------------------------------------------|
| ADMB    | AD Model Builder                           |
| AIC     | Akaike information criterion               |
| AR1     | first-order autoregressive                 |
| ATS     | acoustic-trawl survey                      |
| AVO     | acoustic vessels of opportunity            |
| BTS     | bottom trawl survey                        |
| CIE     | Center for Independent Experts             |
| CPUE    | catch per unit effort                      |
| CSV     | comma-separated values                     |
| DAG     | directed acyclic graph                     |
| DSEM    | dynamic structural equation modeling       |
| ESP     | ecosystem and socioeconomic profile        |
| HTML    | HyperText Markup Language                  |
| IID     | independent and identically distributed    |
| MCMC    | Markov chain Monte Carlo                   |
| MVN     | multivariate normal                        |
| OSA     | one-step-ahead                             |
| RDS     | R serialized data                          |
| RTMB    | R Template Model Builder                   |
| SAFE    | Stock Assessment and Fishery Evaluation    |
| SDNR    | standard deviation of normalized residuals |
| SEM     | structural equation model                  |
| SPM     | Standard Projection Model                  |
| SST     | sea surface temperature                    |

## A-8 Acronyms

This appendix follows the glossary convention in the [Alaska Stock Assessment Report \(asar\)](#) package (Schiano et al. 2026). The project glossary uses ASAR’s `\newacronym` format and adds terms specific to this Rceattle comparison. The quoted label “PM” is a model name.